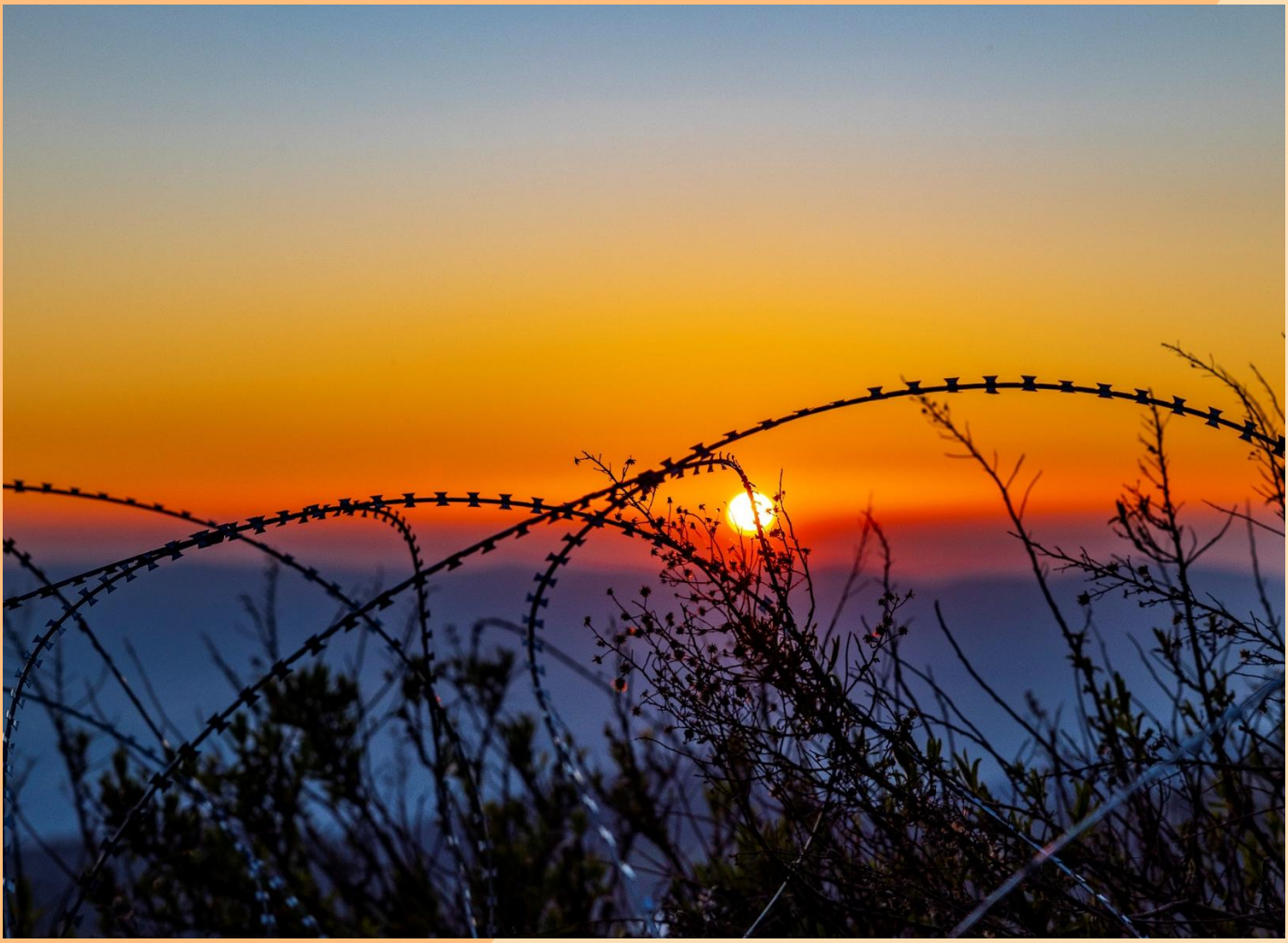

Cybersecurity Academy: Configuring a PA-220 for a Small Office/Home Office

Hunter Zhuang

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Background Information 3
Lab Summary 3
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Purpose

The purpose of this lab is to configure a PA-220 firewall with a small office/home office (SOHO) configuration. In general, SOHO firewalls will have to be configured to give IP addresses through Dynamic Host Configuration Protocol (DHCP) and translate addresses through Network Address Translation (NAT) in addition to filtering unsafe traffic. For the purposes of this lab, simultaneous internet connection and management of the firewall will be focused on, rather than filtering.

Background Information

Firewalls configured for small offices or home offices are important because they prevent attacks on the business. One viable way to have the firewall protect a company network is to just have it placed behind the company router. This way it allows for the firewall to inspect all traffic that goes through it. But this may not be the best location; another common location is to have the company router be behind the firewall.

Furthermore, having the PA-220 route for an office allows it to filter traffic differently from different VLANs or User-ID. This brings significant benefits in that it allows the PA-220 to do authentication, authorization, and accounting for the office network, ultimately keeping things secure.

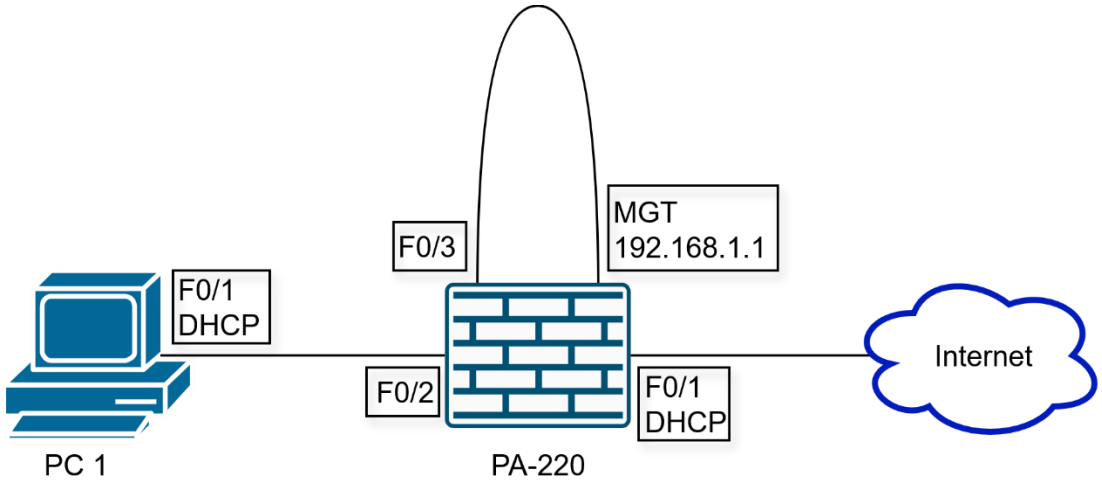
Some companies may need to have public facing servers (like web servers), while also having a private network in a company. The PA-220 can also logically separate these two using VLANs. One can have the public-facing servers in one VLAN and then have the private network on another VLAN. Another way to separate these two networks is to use different physical ports for the network. With these two options, the PA-220 allows flexibility in how someone wants to configure their firewall and security.

The PA-220, made by The Palo Alto Networks®, is a small NGFW made for small businesses and organizations. It has 8 ethernet ports, two power adapters for redundancy, a console port, and a management port. One interesting feature of the PA-220 is that it can detect hostile data even when the data is encrypted by TLS/SSL by decrypting that data. In addition to that, the PA-220 can integrate with Palo Alto Networks' cloud services. This allows the PA-220 to stop known attacks and malicious URLs from affecting devices on the network. There are many other notable features that the PA-220 has, but please refer to the PA-220 specs for the rest of them.

Lab Summary

In this lab, administrators will configure the PA-220 for a SOHO network. The PA-220 will do this by using DHCP to assign IP addresses to hosts and translating internal addresses to the internet (NAT). The PA-220 will act like an edge router to carry out the goals of this lab. In order to connect to the internet and manage the PA-220 through the PC, a piggy-back connection will be needed, since the management port cannot be internally routed through the PA-220. There are obvious security concerns of allowing this piggy-back connection, but it is only for testing, so if the admin disconnects the connection after configuring everything will operate securely.

Network Diagram with IP's



Problems

There were a few issues that were run into in this lab. One issue was confusion around where to find the VLAN *interface* configuration tab vs. the VLAN configuration tab. When I went to the VLANs section under network, I couldn't find anywhere to configure a VLAN interface, but after looking at Palo Alto documentation, I was able to find that VLAN interfaces were to be configured under the interfaces section (which makes sense in hindsight).

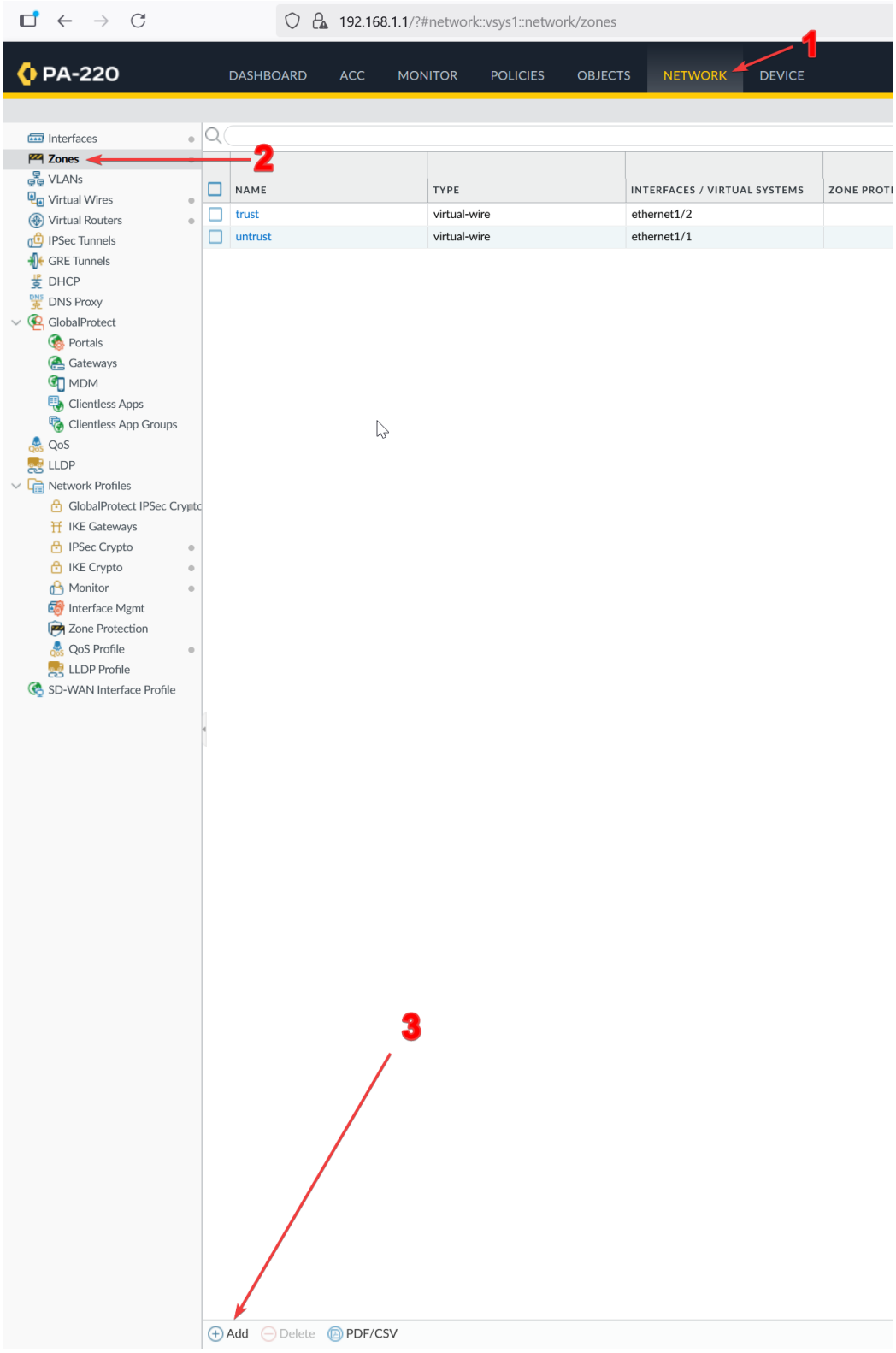
One other issue was the default virtual wire. Since the default virtual wire was no longer assigned to any interfaces after our configuration, the PA-220 was complaining about it when we were trying to commit changes. This issue was easily resolved by deleting the default virtual wire, since there was no need for it.

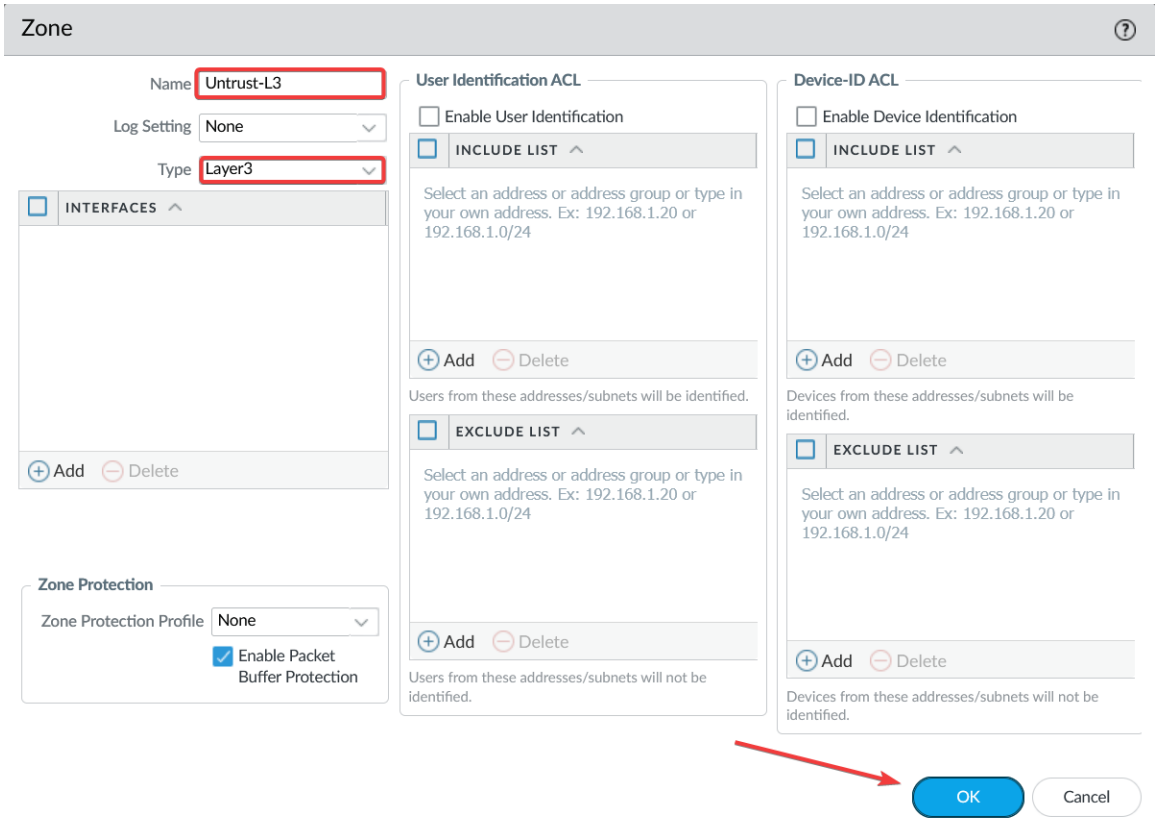
The last issue was a layer 1 issue. Because of the way the lab was set up, the traffic to the internet from our firewall had to go through many of the other internet switches placed around the lab. It turned out that one of the switches in the lab was misconfigured, dropping all of our internet traffic. This issue was really hard to solve, since we thought we had misconfigured the firewall when it was actually a problem with the lab. Later, we found that we couldn't even connect to the internet if directly connected to the internet switch, leading to us fixing the misconfigured internet switch.

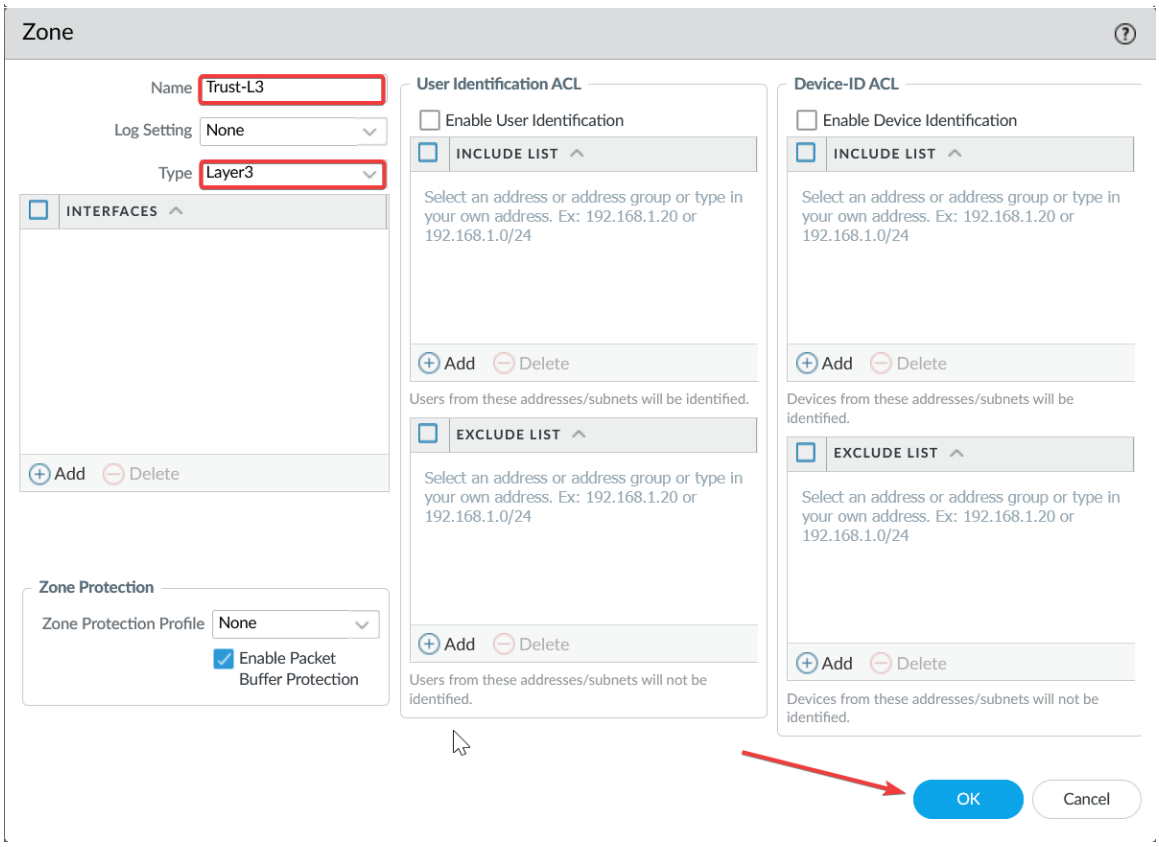
Conclusion

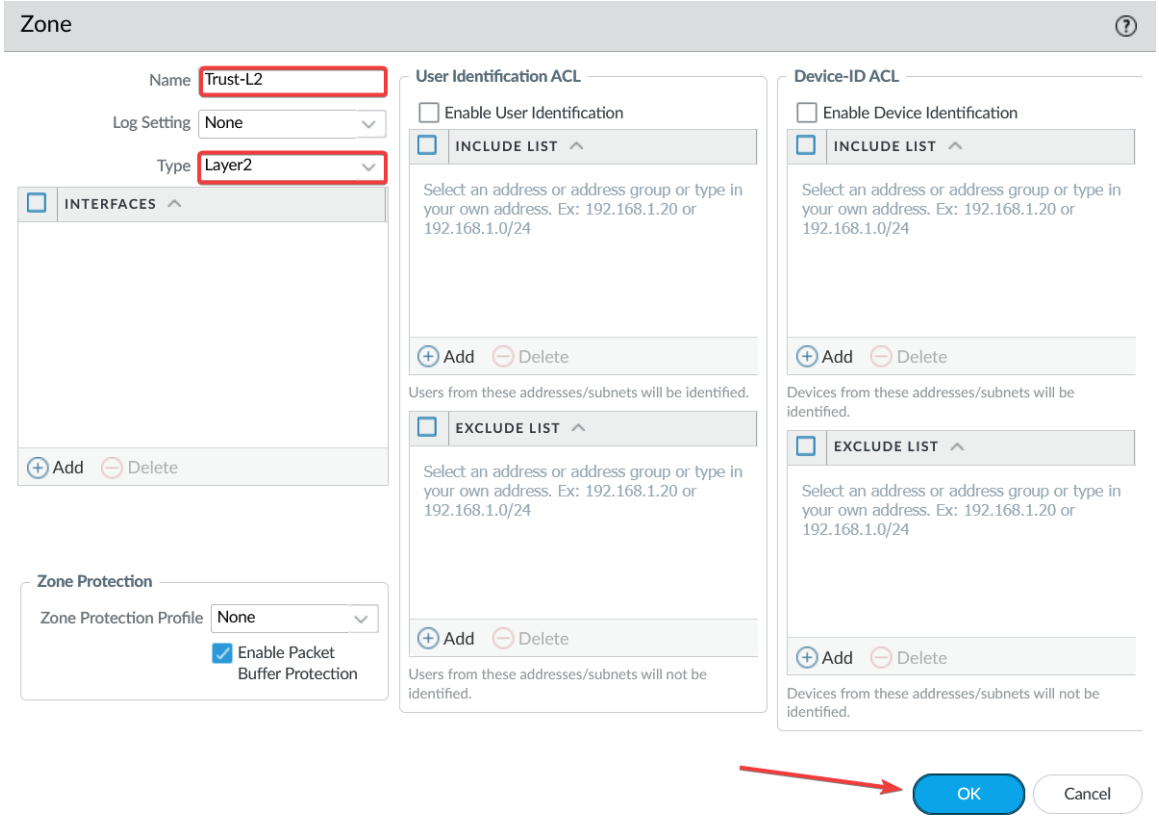
The configuration of this PA-220 is important for any small business because of the need to filter traffic and prevent attacks in a company network. Through the steps outlined in this document any small company can configure their firewall to route devices to the internet, ensuring work goes smoothly for employees. Later, configuration of security policies will be needed to fully protect the company from attacks.

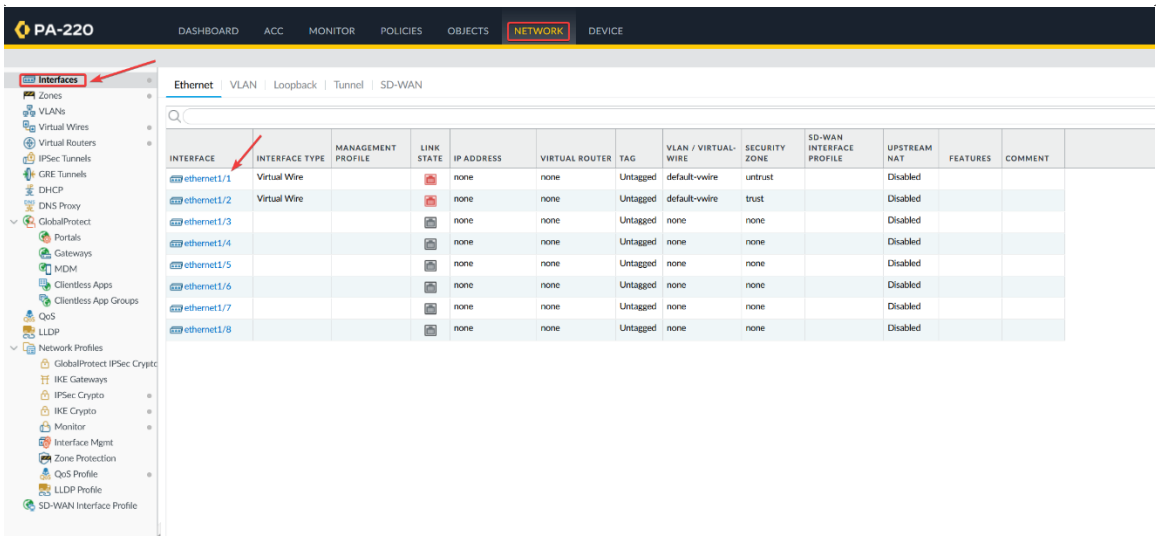
Lab Configuration

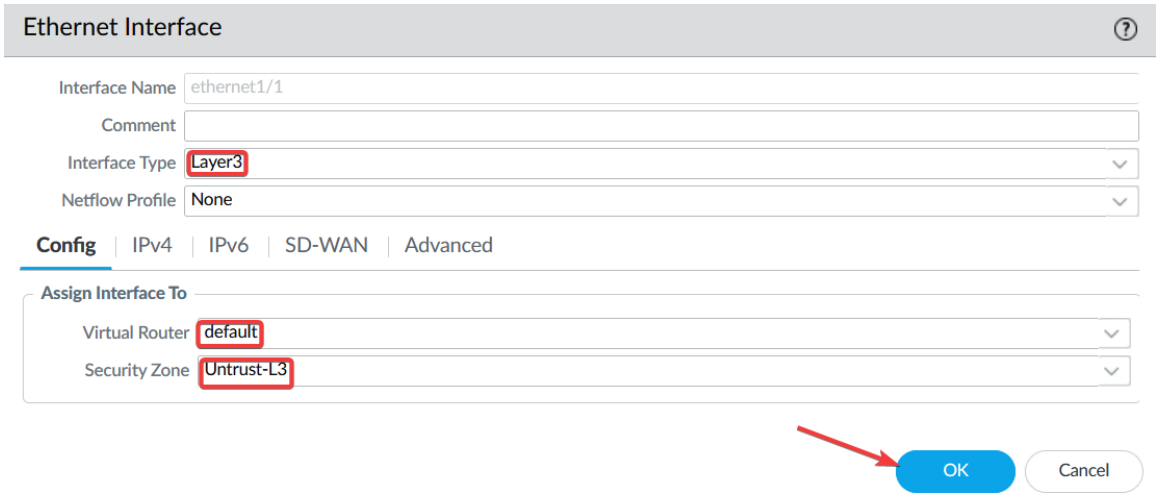


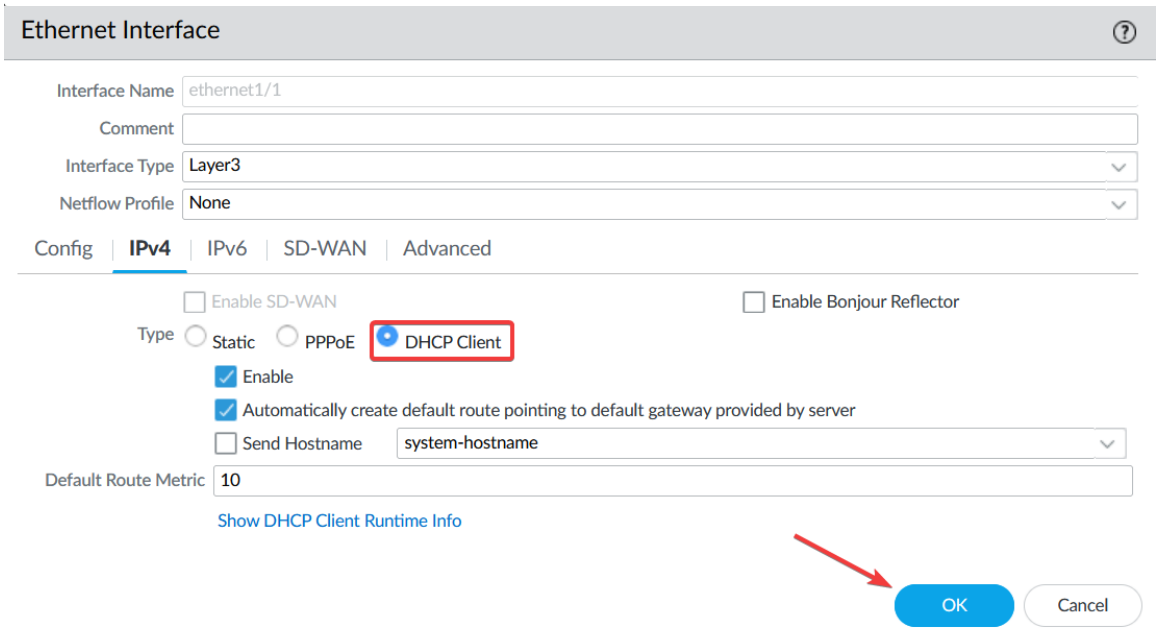

Zone
Name
Untrust-L3
Log Setting
None
Type
Layer3
INTERFACES
Add
Delete
Zone Protection
Zone Protection Profile
None
Enable Packet Buffer Protection
User Identification ACL
Enable User Identification
INCLUDE LIST
Select an address or address group or type in your own address. Ex: 192.168.1.20 or 192.168.1.0/24
Add
Delete
Users from these addresses/subnets will be identified.
EXCLUDE LIST
Select an address or address group or type in your own address. Ex: 192.168.1.20 or 192.168.1.0/24
Add
Delete
Users from these addresses/subnets will not be identified.
Device-ID ACL
Enable Device Identification
INCLUDE LIST
Select an address or address group or type in your own address. Ex: 192.168.1.20 or 192.168.1.0/24
Add
Delete
Devices from these addresses/subnets will be identified.
EXCLUDE LIST
Select an address or address group or type in your own address. Ex: 192.168.1.20 or 192.168.1.0/24
Add
Delete
Devices from these addresses/subnets will not be identified.
OK
Cancel

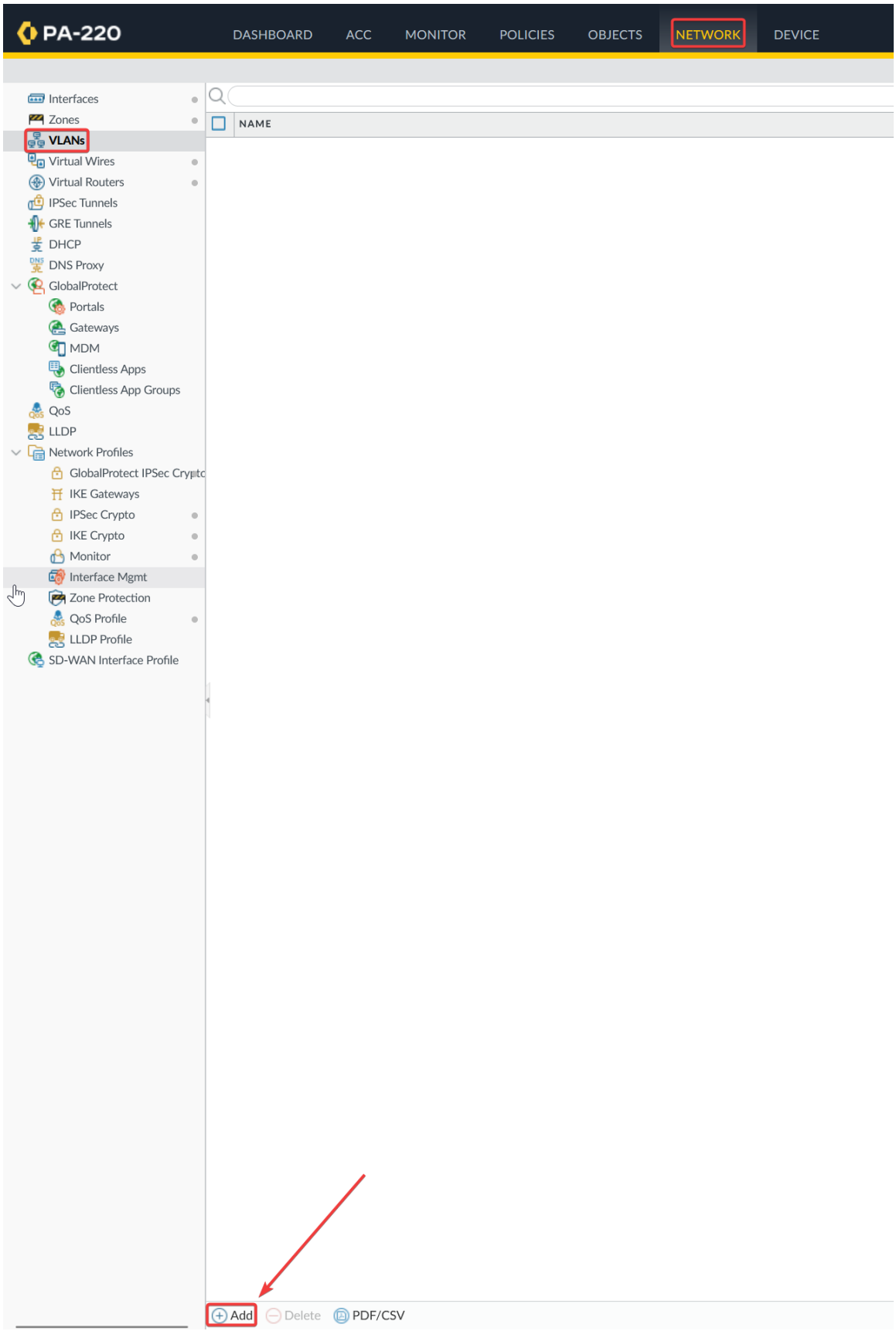

Zone
Name
Trust-L3
Log Setting
None
Type
Layer3
INTERFACES
Add
Delete
Zone Protection
Zone Protection Profile
None
Enable Packet Buffer Protection
User Identification ACL
Enable User Identification
INCLUDE LIST
Select an address or address group or type in your own address. Ex: 192.168.1.20 or 192.168.1.0/24
Add
Delete
Users from these addresses/subnets will be identified.
EXCLUDE LIST
Select an address or address group or type in your own address. Ex: 192.168.1.20 or 192.168.1.0/24
Add
Delete
Users from these addresses/subnets will not be identified.
Device-ID ACL
Enable Device Identification
INCLUDE LIST
Select an address or address group or type in your own address. Ex: 192.168.1.20 or 192.168.1.0/24
Add
Delete
Devices from these addresses/subnets will be identified.
EXCLUDE LIST
Select an address or address group or type in your own address. Ex: 192.168.1.20 or 192.168.1.0/24
Add
Delete
Devices from these addresses/subnets will not be identified.
OK
Cancel

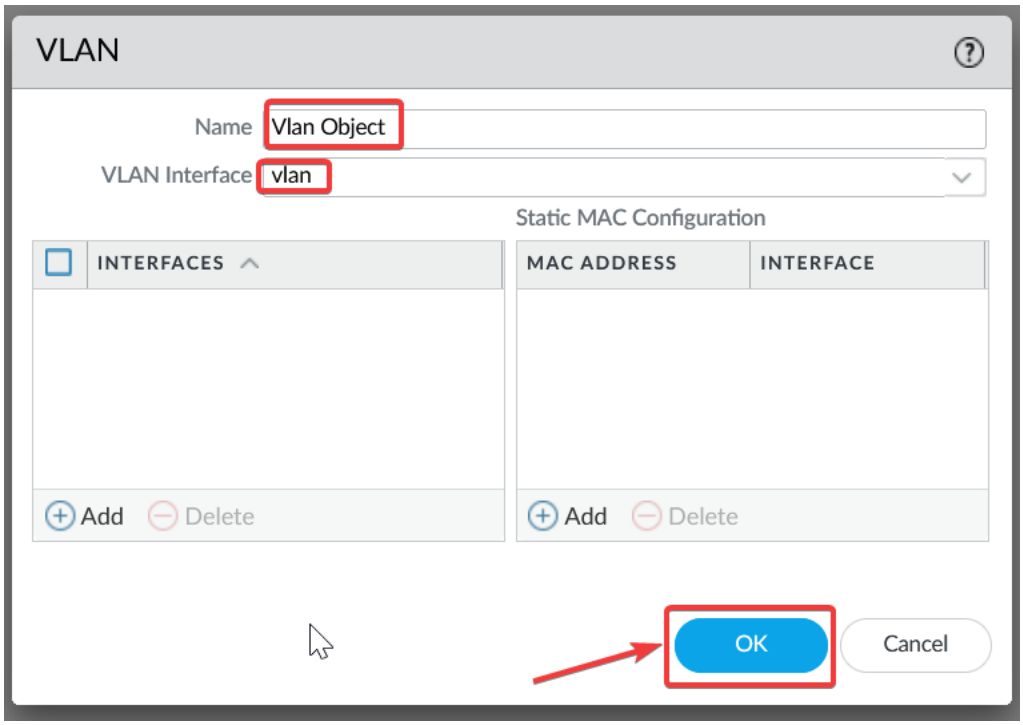

Zone
Name
Trust-L2
Log Setting
None
Type
Layer2
INTERFACES
Add
Delete
Zone Protection
Zone Protection Profile
None
Enable Packet Buffer Protection
User Identification ACL
Enable User Identification
INCLUDE LIST
Select an address or address group or type in your own address. Ex: 192.168.1.20 or 192.168.1.0/24
Add
Delete
Users from these addresses/subnets will be identified.
EXCLUDE LIST
Select an address or address group or type in your own address. Ex: 192.168.1.20 or 192.168.1.0/24
Add
Delete
Users from these addresses/subnets will not be identified.
Device-ID ACL
Enable Device Identification
INCLUDE LIST
Select an address or address group or type in your own address. Ex: 192.168.1.20 or 192.168.1.0/24
Add
Delete
Devices from these addresses/subnets will be identified.
EXCLUDE LIST
Select an address or address group or type in your own address. Ex: 192.168.1.20 or 192.168.1.0/24
Add
Delete
Devices from these addresses/subnets will not be identified.
OK
Cancel

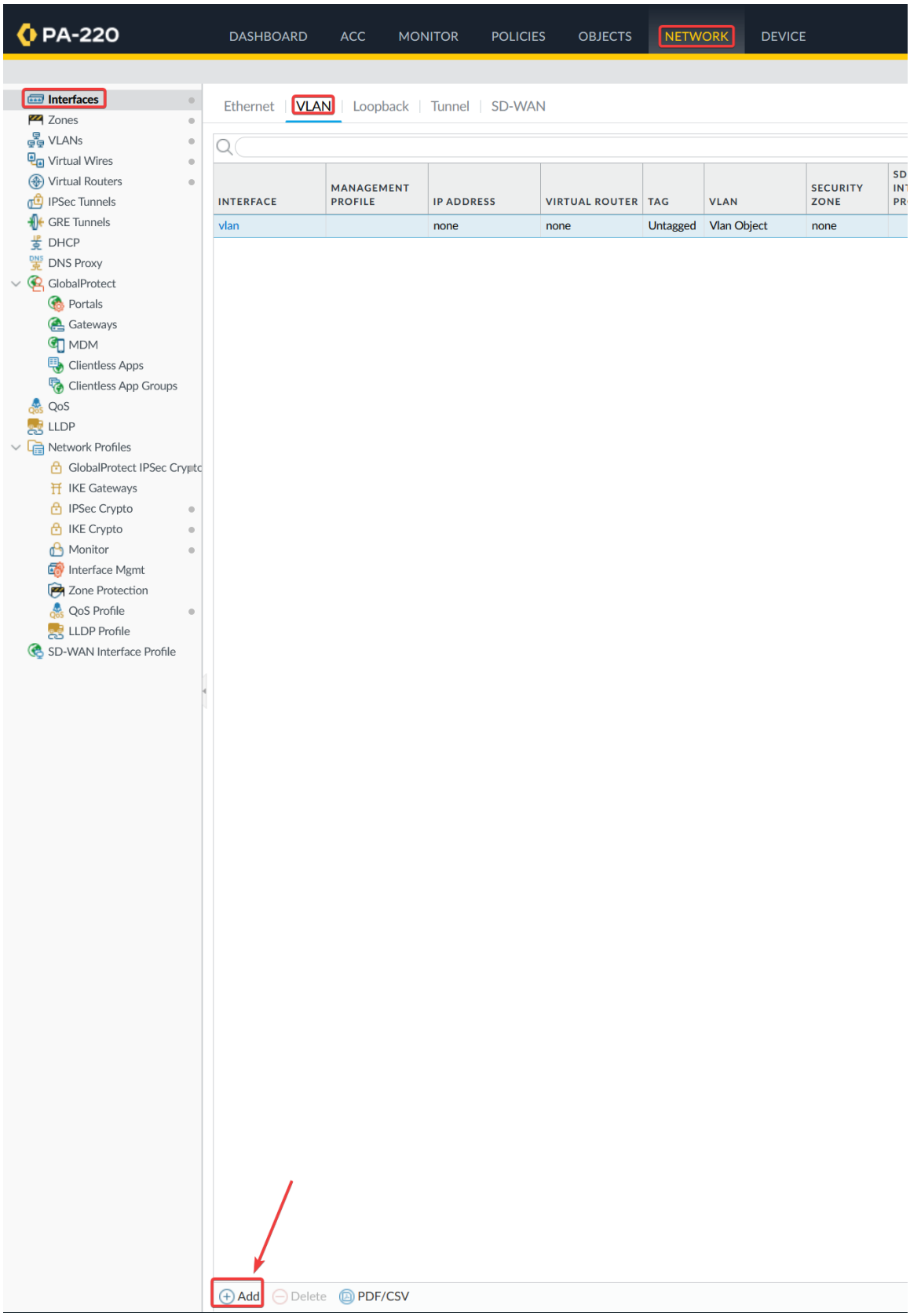

PA-220
DASHBOARD
ACC
MONITOR
POLICIES
OBJECTS
NETWORK
DEVICE
Interfaces
Zones
VLANs
Virtual Wires
Virtual Routers
IPSec Tunnels
GRE Tunnels
DHCP
DNS Proxy
GlobalProtect
Portals
Gateways
MDM
Clientless Apps
Clientless App Groups
QoS
LLDP
Network Profiles
GlobalProtect IPSec Crypto
IKE Gateways
IPSec Crypto
IKE Crypto
Monitor
Interface Mgmt
Zone Protection
QoS Profile
LLDP Profile
SD-WAN Interface Profile
Ethernet
VLAN
Loopback
Tunnel
SD-WAN
INTERFACE
INTERFACE TYPE
MANAGEMENT PROFILE
LINK STATE
IP ADDRESS
VIRTUAL ROUTER
TAG
VLAN / VIRTUAL-WIRE
SECURITY ZONE
SD-WAN INTERFACE PROFILE
UPSTREAM NAT
FEATURES
COMMENT
ethernet1/1 Virtual Wire none none Untagged default-vwire untrust Disabled
ethernet1/2 Virtual Wire none none Untagged default-vwire trust Disabled
ethernet1/3 none none Untagged none none Disabled
ethernet1/4 none none Untagged none none Disabled
ethernet1/5 none none Untagged none none Disabled
ethernet1/6 none none Untagged none none Disabled
ethernet1/7 none none Untagged none none Disabled
ethernet1/8 none none Untagged none none Disabled


Ethernet Interface
Interface Name
ethernet1/1
Comment
Interface Type
Layer3
Netflow Profile
None
Config
IPv4
IPv6
SD-WAN
Advanced
Assign Interface To
Virtual Router
default
Security Zone
Untrust-L3
OK
Cancel


Ethernet Interface
Interface Name
ethernet1/1
Comment
Interface Type
Layer3
Netflow Profile
None
Config
IPv4
IPv6
SD-WAN
Advanced
Enable SD-WAN
Enable Bonjour Reflector
Type
Static
PPPoE
DHCP Client
Enable
Automatically create default route pointing to default gateway provided by server
Send Hostname
system-hostname
Default Route Metric
10
Show DHCP Client Runtime Info
OK
Cancel


PA-220
DASHBOARD
ACC
MONITOR
POLICIES
OBJECTS
NETWORK
DEVICE
Interfaces
Zones
VLANs
Virtual Wires
Virtual Routers
IPSec Tunnels
GRE Tunnels
DHCP
DNS Proxy
GlobalProtect
Portals
Gateways
MDM
Clientless Apps
Clientless App Groups
QoS
LLDP
Network Profiles
GlobalProtect IPSec Crypto
IKE Gateways
IPSec Crypto
IKE Crypto
Monitor
Interface Mgmt
Zone Protection
QoS Profile
LLDP Profile
SD-WAN Interface Profile
NAME
Add
Delete
PDF/CSV


VLAN
Name
Vlan Object
VLAN Interface
vlan
Static MAC Configuration
INTERFACES
MAC ADDRESS
INTERFACE
Add
Delete
Add
Delete
OK
Cancel


PA-220
DASHBOARD
ACC
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POLICIES
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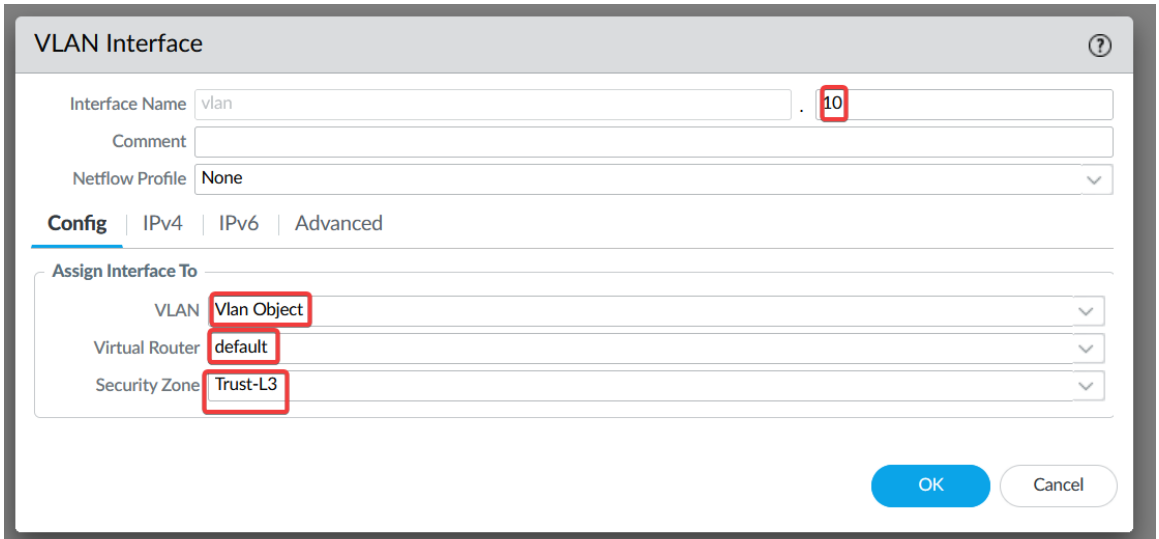
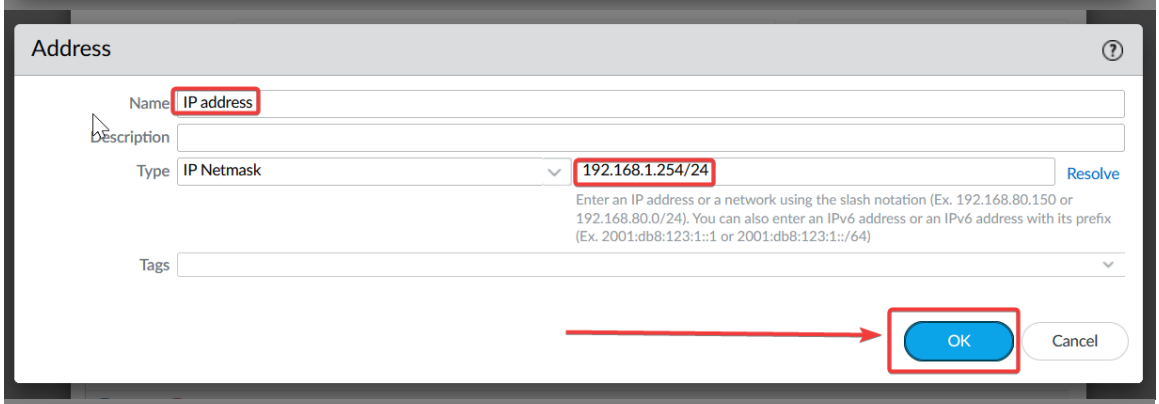
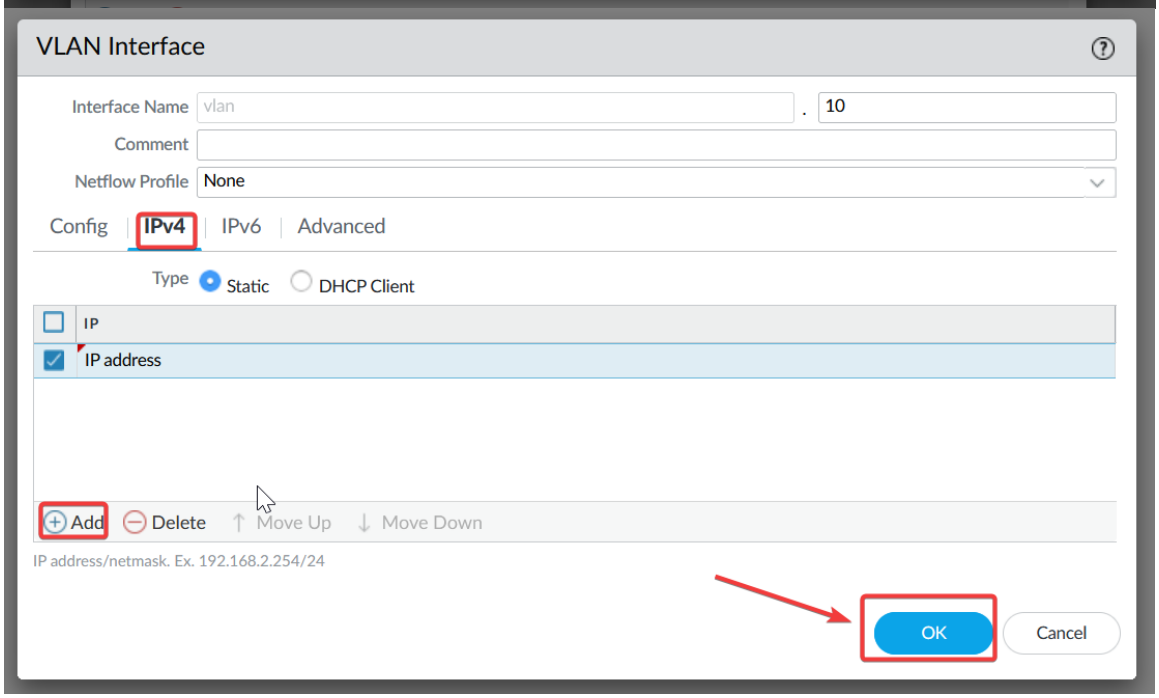
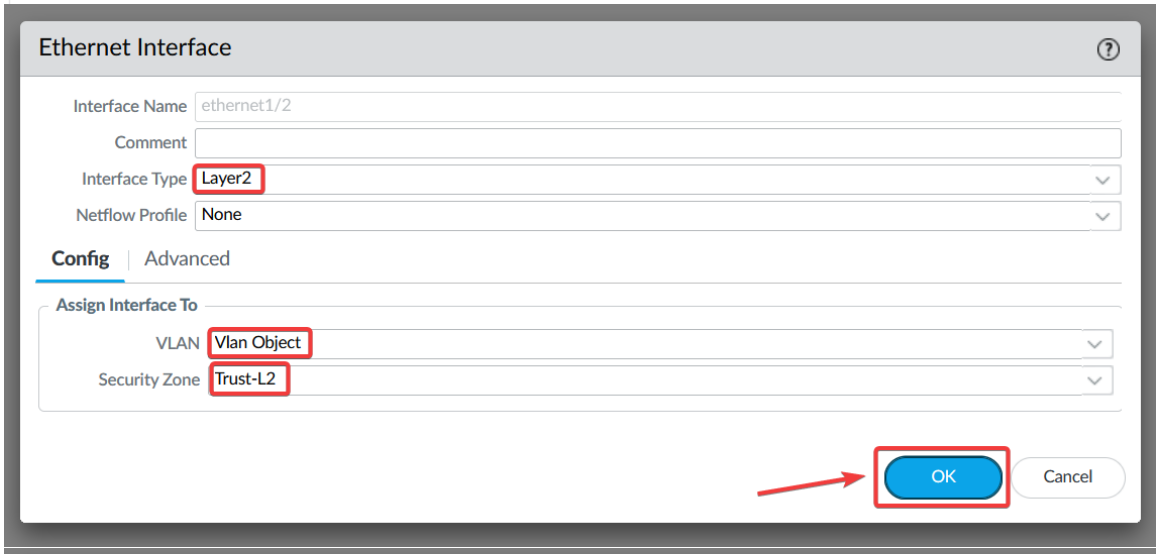
NETWORK
DEVICE
Interfaces
Zones
VLANs
Virtual Wires
Virtual Routers
IPSec Tunnels
GRE Tunnels
DHCP
DNS Proxy
GlobalProtect
Portals
Gateways
MDM
Clientless Apps
Clientless App Groups
QoS
LLDP
Network Profiles
GlobalProtect IPSec Crypto
IKE Gateways
IPSec Crypto
IKE Crypto
Monitor
Interface Mgmt
Zone Protection
QoS Profile
LLDP Profile
SD-WAN Interface Profile
Ethernet
VLAN
Loopback
Tunnel
SD-WAN
INTERFACE
MANAGEMENT PROFILE
IP ADDRESS
VIRTUAL ROUTER
TAG
VLAN
SECURITY ZONE
vlan
none
none
Untagged
Vlan Object
none
Add
Delete
PDF/CSV

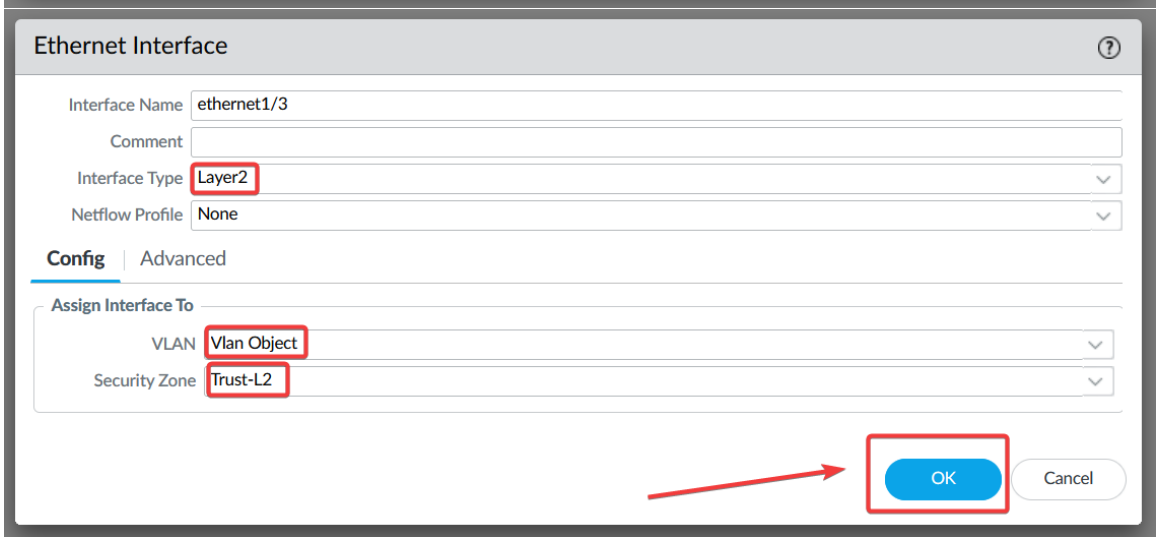

VLAN Interface
Interface Name
vlan
10
Comment
Netflow Profile
None
Config
IPv4
IPv6
Advanced
Assign Interface To
VLAN
Vlan Object
Virtual Router
default
Security Zone
Trust-L3
OK
Cancel

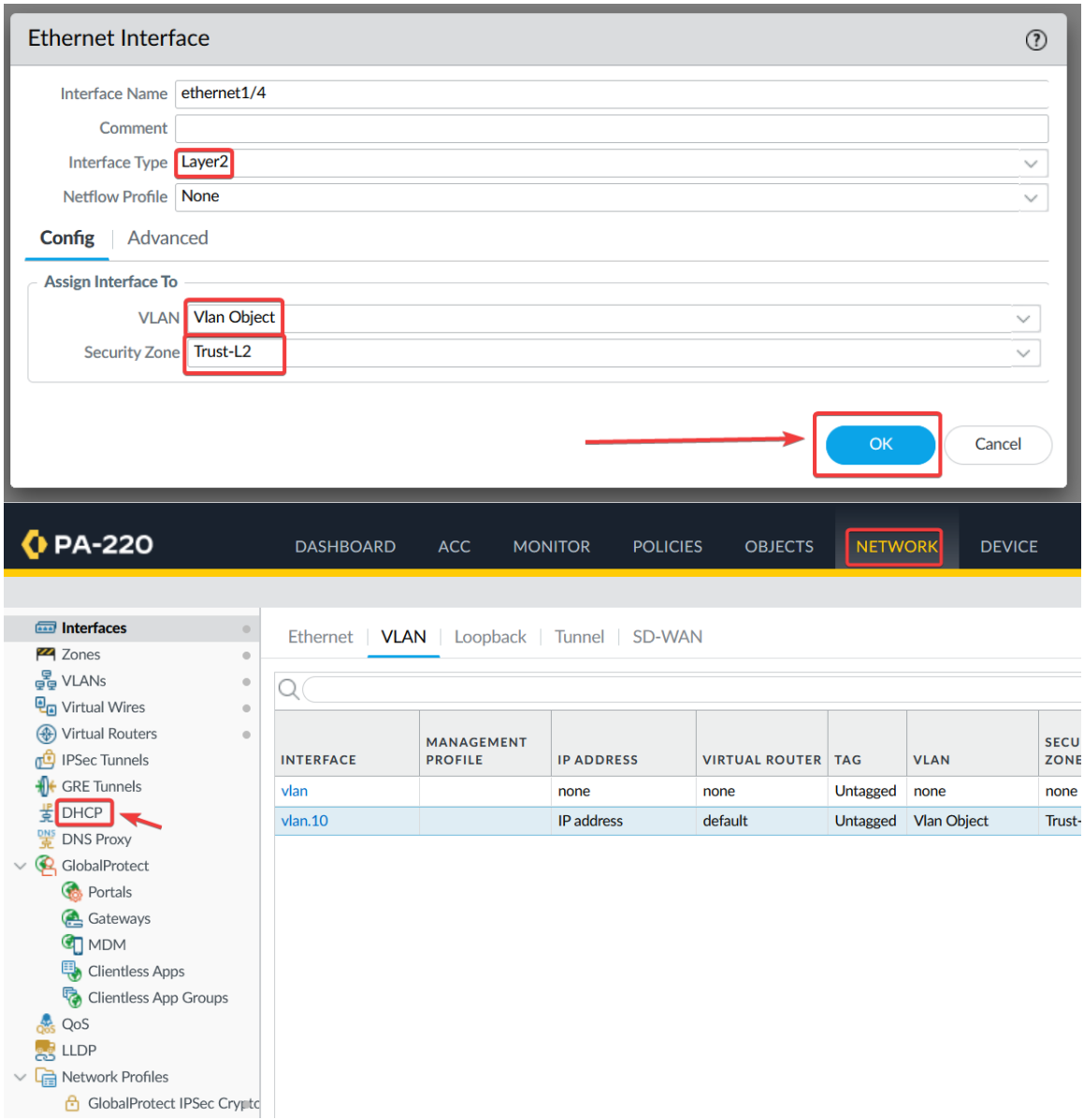

Address
Name
IP address
Description
Type
IP Netmask
192.168.1.254/24
Resolve
Enter an IP address or a network using the slash notation (Ex. 192.168.80.150 or 192.168.80.0/24). You can also enter an IPv6 address or an IPv6 address with its prefix (Ex. 2001:db8:123:1::1 or 2001:db8:123:1::/64)
Tags
OK
Cancel

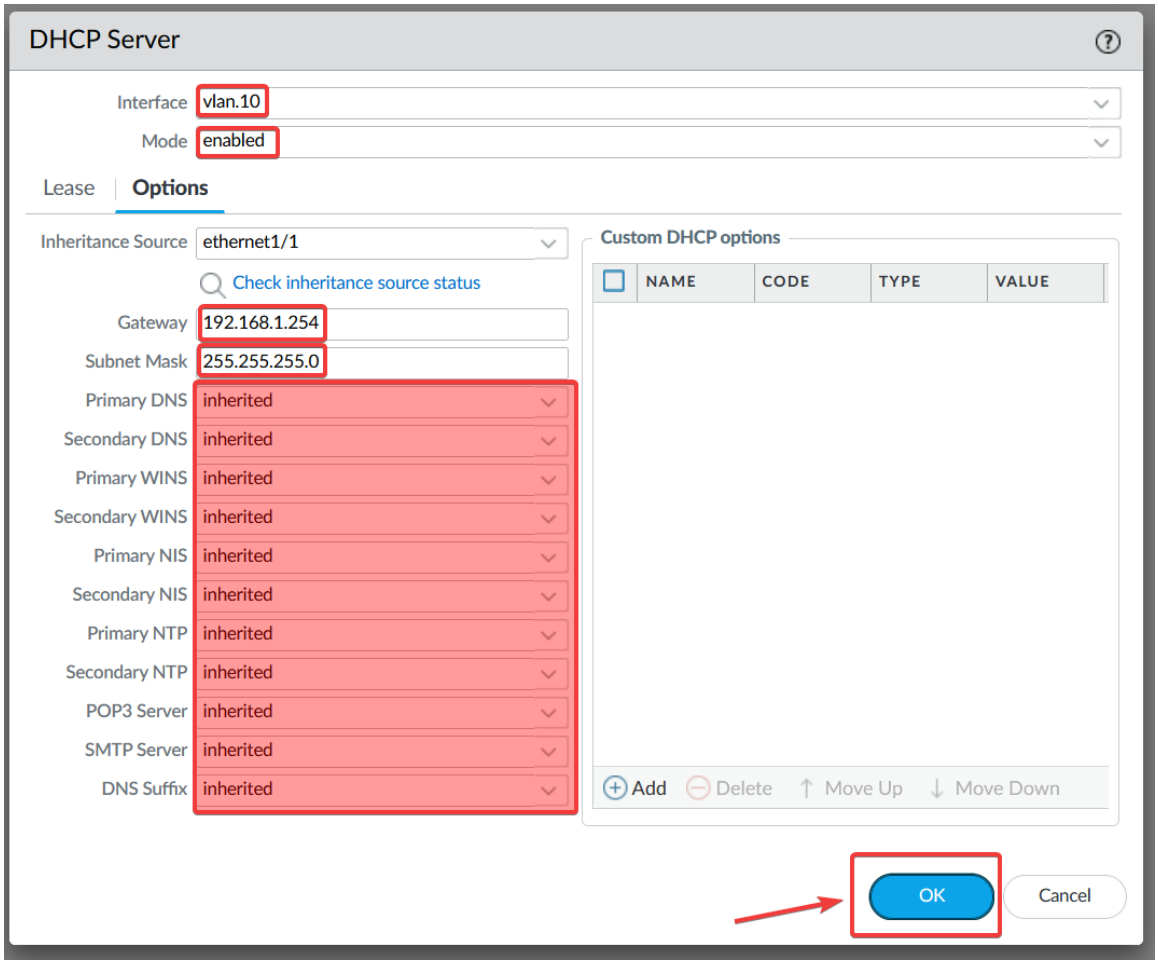

VLAN Interface
Interface Name
vlan
10
Comment
Netflow Profile
None
Config
IPv4
IPv6
Advanced
Type
Static
DHCP Client
IP
IP address
Add
Delete
Move Up
Move Down
IP address/netmask. Ex. 192.168.2.254/24
OK
Cancel

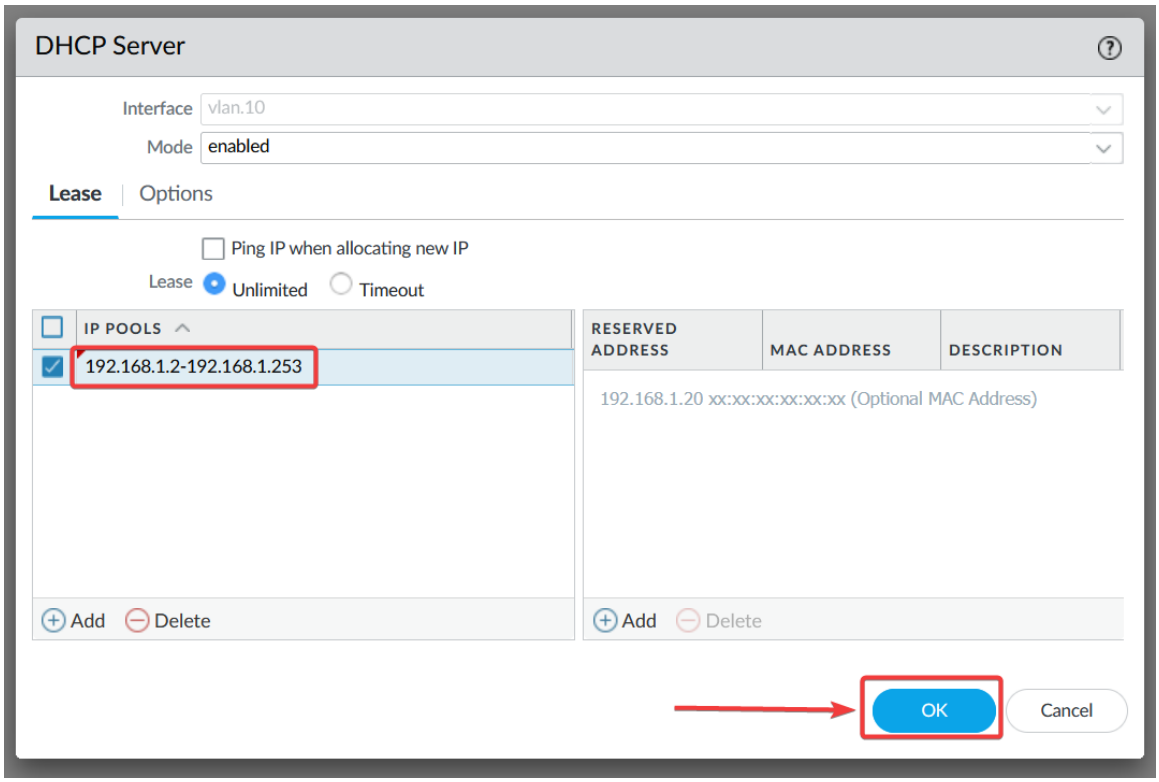
INTERFACE	INTERFACE TYPE	MANAGEMENT PROFILE	LINK STATE	IP ADDRESS	VIRTUAL ROUTER	TAG	VLAN / VIRTUAL-WIRE	SECURITY ZONE	SD-WAN INTERFACE PROFILE	UPSTREAM NAT	FEATURES	COMMENT
ethernet1/1	Layer3			Dynamic-DHCP Client	default	Untagged	none	Untrust-L3		Disabled		
ethernet1/2	Virtual Wire			none	none	Untagged	none	none		Disabled		
ethernet1/3				none	none	Untagged	none	none		Disabled		
ethernet1/4				none	none	Untagged	none	none		Disabled		
ethernet1/5				none	none	Untagged	none	none		Disabled		
ethernet1/6				none	none	Untagged	none	none		Disabled		
ethernet1/7				none	none	Untagged	none	none		Disabled		
ethernet1/8				none	none	Untagged	none	none		Disabled		

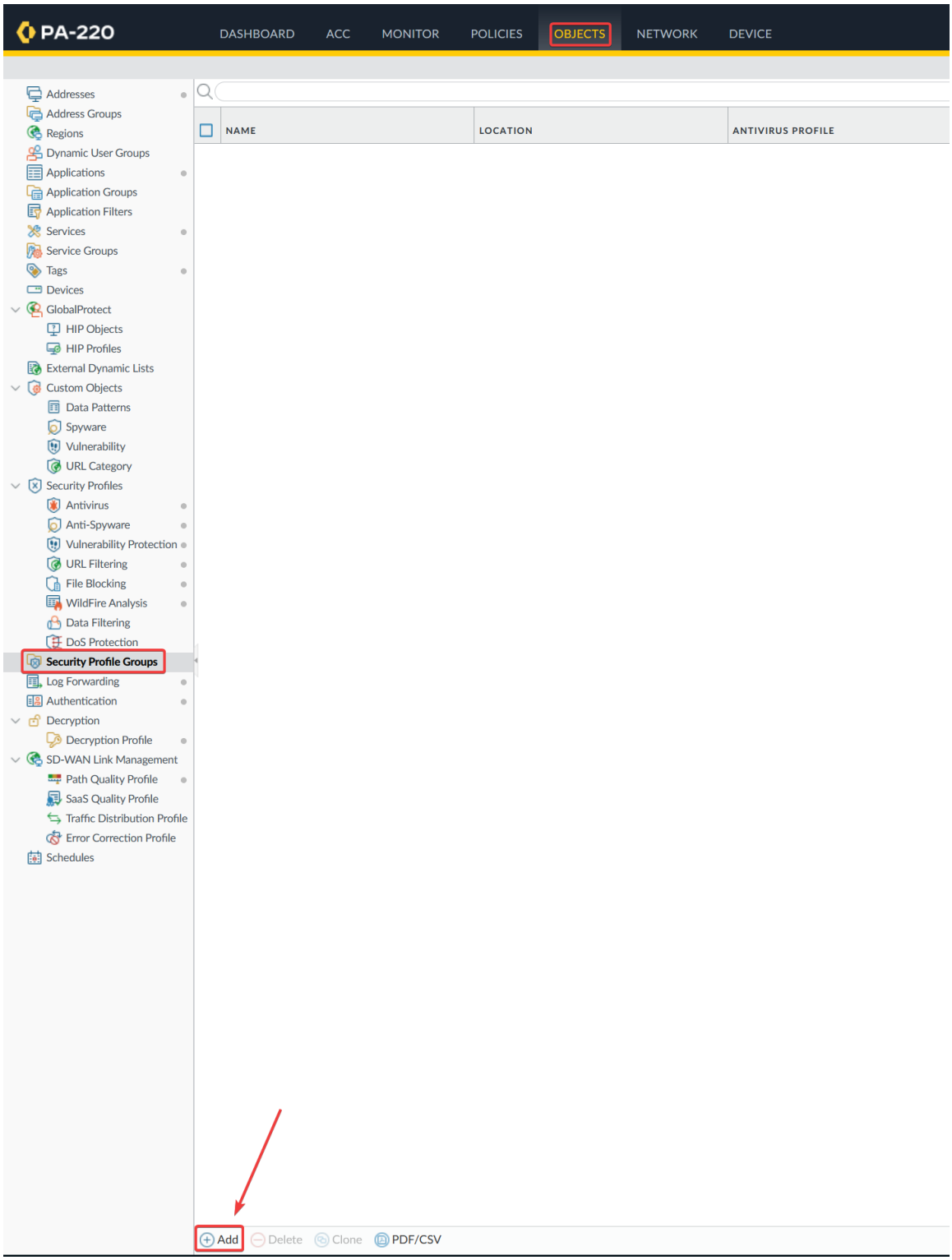


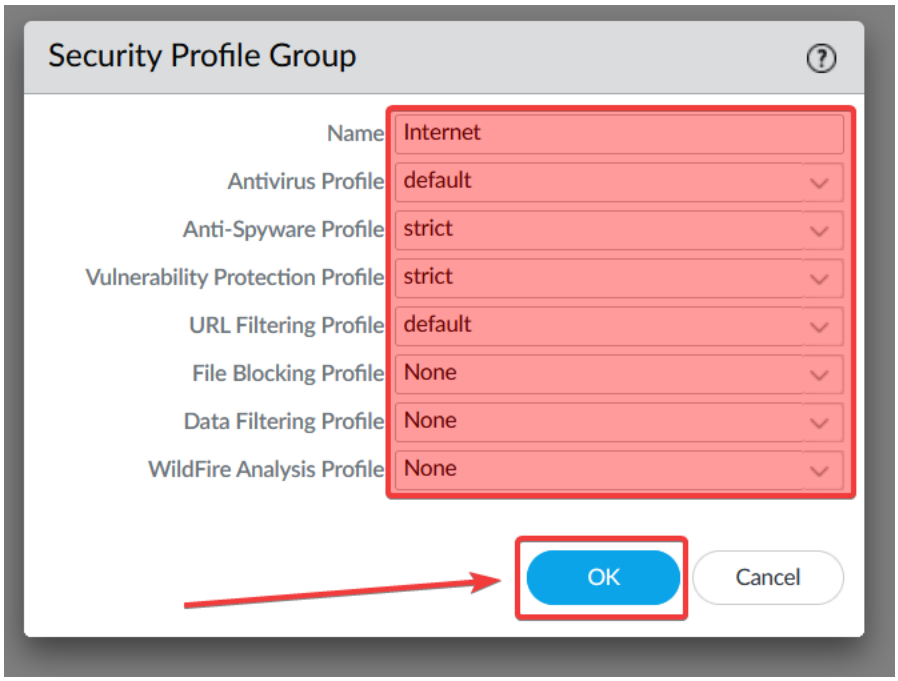



Ethernet Interface
Interface Name
ethernet1/4
Comment
Interface Type
Layer2
Netflow Profile
None
Config
Advanced
Assign Interface To
VLAN
Vlan Object
Security Zone
Trust-L2
OK
Cancel
PA-220
DASHBOARD
ACC
MONITOR
POLICIES
OBJECTS
NETWORK
DEVICE
Interfaces
Zones
VLANs
Virtual Wires
Virtual Routers
IPSec Tunnels
GRE Tunnels
DHCP
DNS Proxy
GlobalProtect
Portals
Gateways
MDM
Clientless Apps
Clientless App Groups
QoS
LLDP
Network Profiles
GlobalProtect IPSec Crypto
Ethernet
VLAN
Loopback
Tunnel
SD-WAN
INTERFACE
MANAGEMENT PROFILE
IP ADDRESS
VIRTUAL ROUTER
TAG
VLAN
SECU ZONE
vlan
none
none
Untagged
none
none
vlan.10
IP address
default
Untagged
Vlan Object
Trust-


DHCP Server
Interface
vlan.10
Mode
enabled
Lease
Options
Inheritance Source
ethernet1/1
Check inheritance source status
Gateway
192.168.1.254
Subnet Mask
255.255.255.0
Primary DNS
inherited
Secondary DNS
inherited
Primary WINS
inherited
Secondary WINS
inherited
Primary NIS
inherited
Secondary NIS
inherited
Primary NTP
inherited
Secondary NTP
inherited
POP3 Server
inherited
SMTP Server
inherited
DNS Suffix
inherited
Custom DHCP options
NAME
CODE
TYPE
VALUE
Add
Delete
Move Up
Move Down
OK
Cancel


DHCP Server
Interface
vlan.10
Mode
enabled
Lease
Options
Ping IP when allocating new IP
Lease
Unlimited
Timeout
IP POOLS
192.168.1.2-192.168.1.253
RESERVED ADDRESS
MAC ADDRESS
DESCRIPTION
192.168.1.20 xx:xx:xx:xx:xx:xx (Optional MAC Address)
Add
Delete
Add
Delete
OK
Cancel


PA-220
DASHBOARD
ACC
MONITOR
POLICIES
OBJECTS
NETWORK
DEVICE
Addresses
Address Groups
Regions
Dynamic User Groups
Applications
Application Groups
Application Filters
Services
Service Groups
Tags
Devices
GlobalProtect
HIP Objects
HIP Profiles
External Dynamic Lists
Custom Objects
Data Patterns
Spyware
Vulnerability
URL Category
Security Profiles
Antivirus
Anti-Spyware
Vulnerability Protection
URL Filtering
File Blocking
WildFire Analysis
Data Filtering
DoS Protection
Security Profile Groups
Log Forwarding
Authentication
Decryption
Decryption Profile
SD-WAN Link Management
Path Quality Profile
SaaS Quality Profile
Traffic Distribution Profile
Error Correction Profile
Schedules
NAME
LOCATION
ANTIVIRUS PROFILE
Add
Delete
Clone
PDF/CSV


Security Profile Group
Name
Internet
Antivirus Profile
default
Anti-Spyware Profile
strict
Vulnerability Protection Profile
strict
URL Filtering Profile
default
File Blocking Profile
None
Data Filtering Profile
None
WildFire Analysis Profile
None
OK
Cancel

PA-220

DASHBOARD | ACC | MONITOR | POLICIES | OBJECTS | NETWORK | DEVICE

- Security
- NAT
- QoS
- Policy Based Forwarding
- Decryption
- Tunnel Inspection
- Application Override
- Authentication
- DoS Protection
- SD-WAN

				Source				Destination			
	NAME	TAGS	TYPE	ZONE	ADDRESS	USER	DEVICE	ZONE	ADDRESS	DEVICE	APPLIC
1	rule1	none	universal	trust	any	any	any	untrust	any	any	any
2	intrazone-default	none	intrazone	any	any	any	any	(intrazone)	any	any	any
3	interzone-default	none	interzone	any	any	any	any	any	any	any	any

Policy Optimizer

- New App Viewer
- Rules Without App Controls
- Unused Apps
- Log Forwarding for Security Ser
- Rule Usage
  - Unused in 30 days
  - Unused in 90 days
  - Unused

Tag Browser

Object : Addresses

Add | Delete | Clone | Override | Revert | Enable | Disable | Move | PDF/CSV | Highlight Unused Rules | Default View | Reset Rule Hit Counter | Group | Tes

Security Policy Rule

General | Source | Destination | Application | Service/URL Category | Actions

Name: Internet Outgoing

Rule Type: universal (default)

Description: All traffic to the internet.

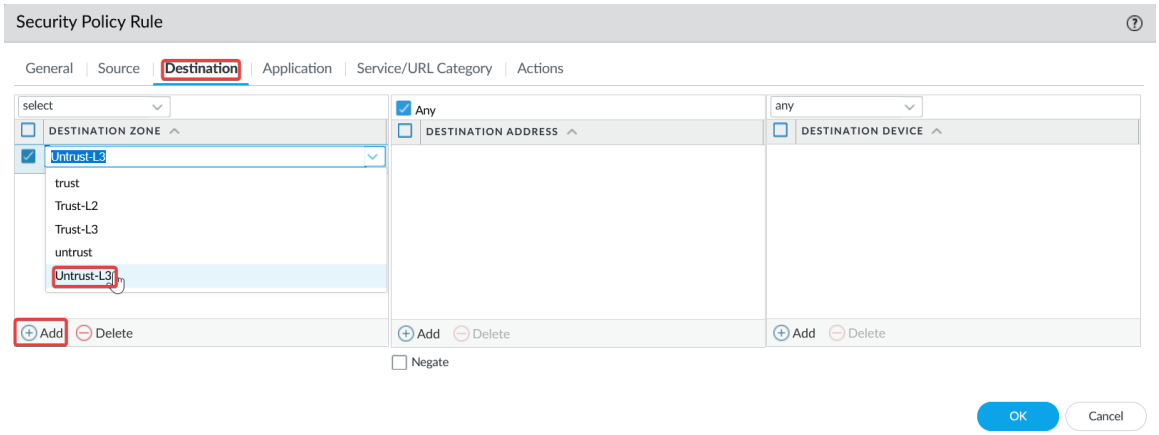
Tags:

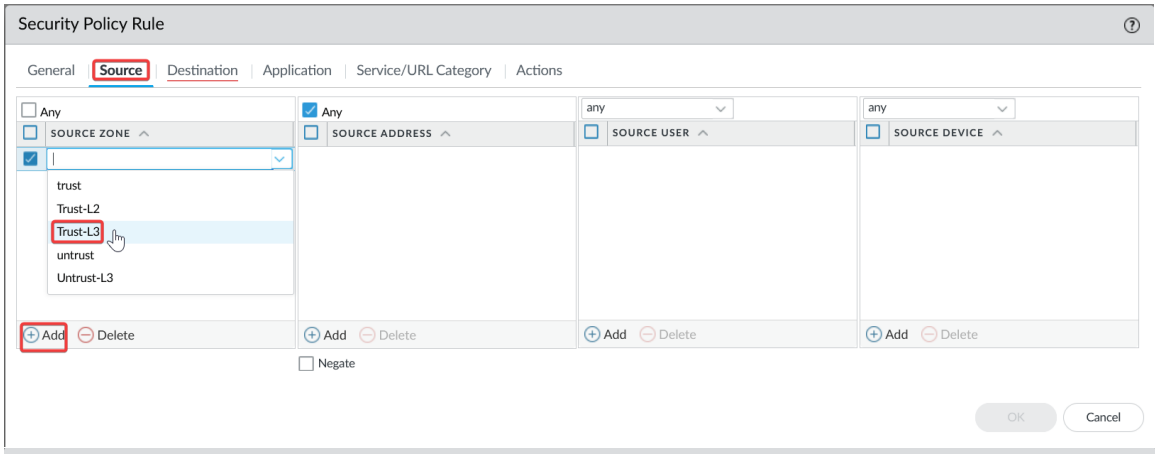
Group Rules By Tag: None

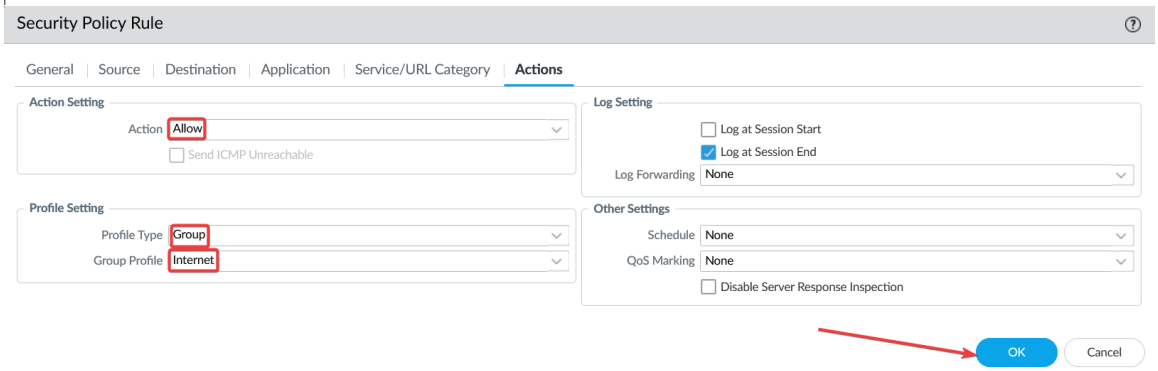
Audit Comment:

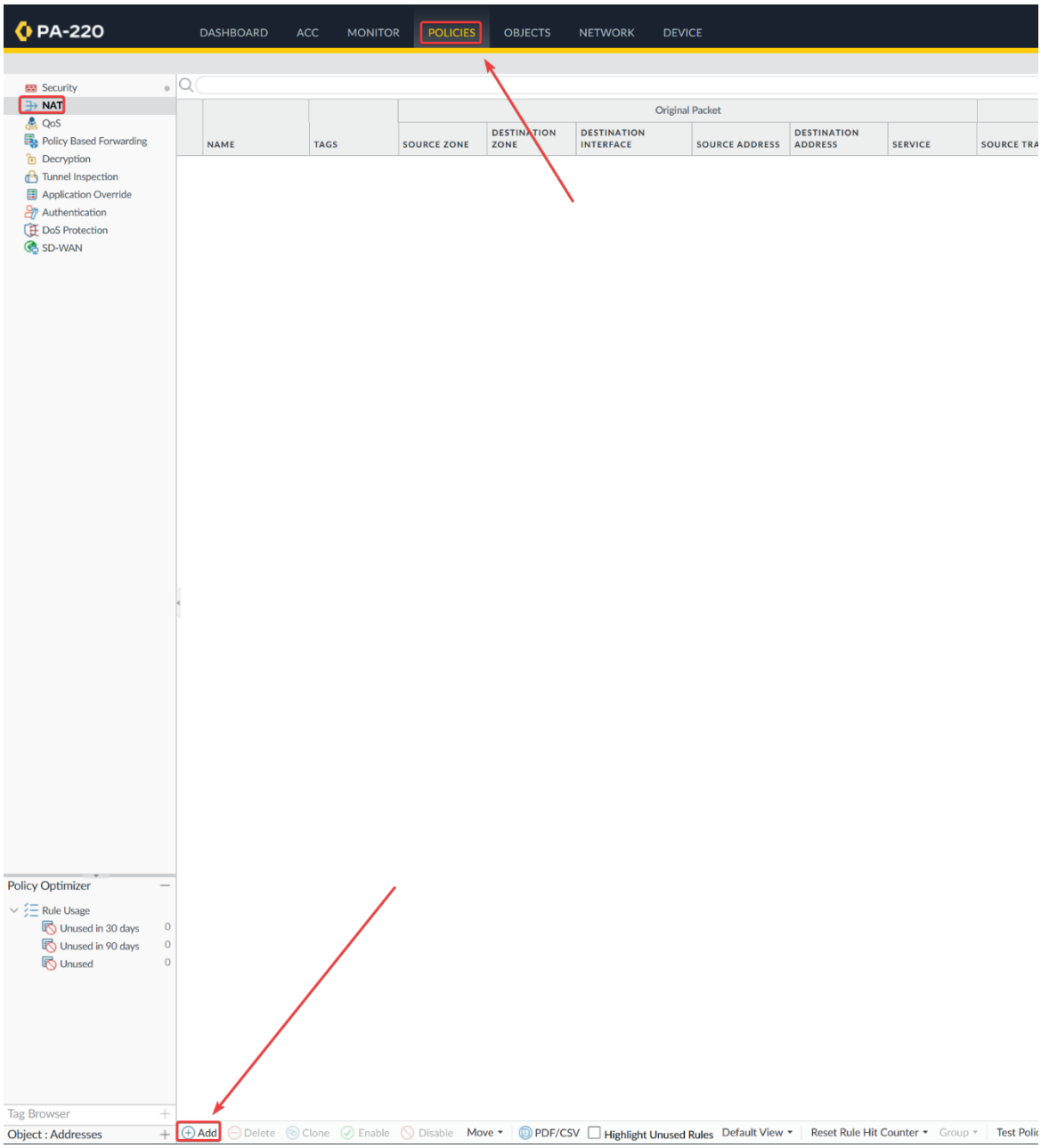
Audit Comment Archive

OK | Cancel


Security Policy Rule
General
Source
Destination
Application
Service/URL Category
Actions
select
DESTINATION ZONE
Untrust-L3
trust
Trust-L2
Trust-L3
untrust
Untrust-L3
Add
Delete
Any
DESTINATION ADDRESS
Add
Delete
any
DESTINATION DEVICE
Add
Delete
Negate
OK
Cancel


Security Policy Rule
General
Source
Destination
Application
Service/URL Category
Actions
Any
SOURCE ZONE
trust
Trust-L2
Trust-L3
untrust
Untrust-L3
Add
Delete
Any
SOURCE ADDRESS
Add
Delete
any
SOURCE USER
Add
Delete
any
SOURCE DEVICE
Add
Delete
Negate
OK
Cancel


Security Policy Rule
General
Source
Destination
Application
Service/URL Category
Actions
Action Setting
Action
Allow
Send ICMP Unreachable
Log Setting
Log at Session Start
Log at Session End
Log Forwarding
None
Profile Setting
Profile Type
Group
Group Profile
Internet
Other Settings
Schedule
None
QoS Marking
None
Disable Server Response Inspection
OK
Cancel


PA-220
DASHBOARD
ACC
MONITOR
POLICIES
OBJECTS
NETWORK
DEVICE
Security
NAT
QoS
Policy Based Forwarding
Decryption
Tunnel Inspection
Application Override
Authentication
DoS Protection
SD-WAN
Original Packet
NAME
TAGS
SOURCE ZONE
DESTINATION ZONE
DESTINATION INTERFACE
SOURCE ADDRESS
DESTINATION ADDRESS
SERVICE
Policy Optimizer
Rule Usage
Unused in 30 days 0
Unused in 90 days 0
Unused 0
Tag Browser
Object : Addresses
Add
Delete
Clone
Enable
Disable
Move
PDF/CSV
Highlight Unused Rules
Default View
Reset Rule Hit Counter
Group

NAT Policy Rule

General | Original Packet | Translated Packet

Name: Internet Outgoing

Description:

Tags:

Group Rules By Tag: None

NAT Type: ipv4

Audit Comment:

Audit Comment Archive

OK | Cancel

NAT Policy Rule

General | Original Packet | Translated Packet

Any

SOURCE ZONE

Trust-L3

Add | Delete

Destination Zone: Untrust-L3

Destination Interface: ethernet1/1

Service: any

Any

SOURCE ADDRESS

Add | Delete

Any

DESTINATION ADDRESS

Add | Delete

OK | Cancel

NAT Policy Rule

General | Original Packet | Translated Packet

Source Address Translation

Translation Type: Dynamic IP And Port

Address Type: Interface Address

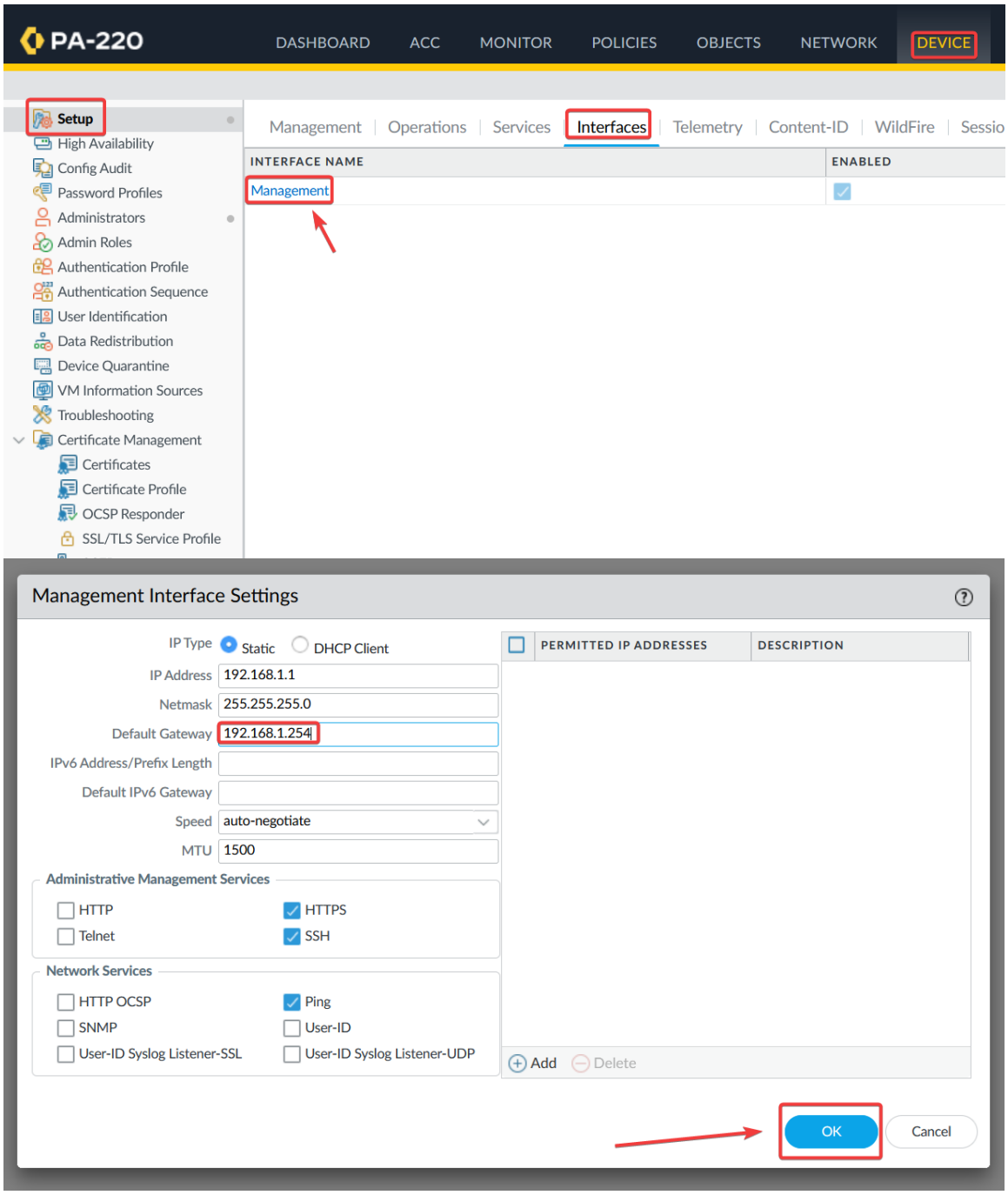
Interface: ethernet1/1

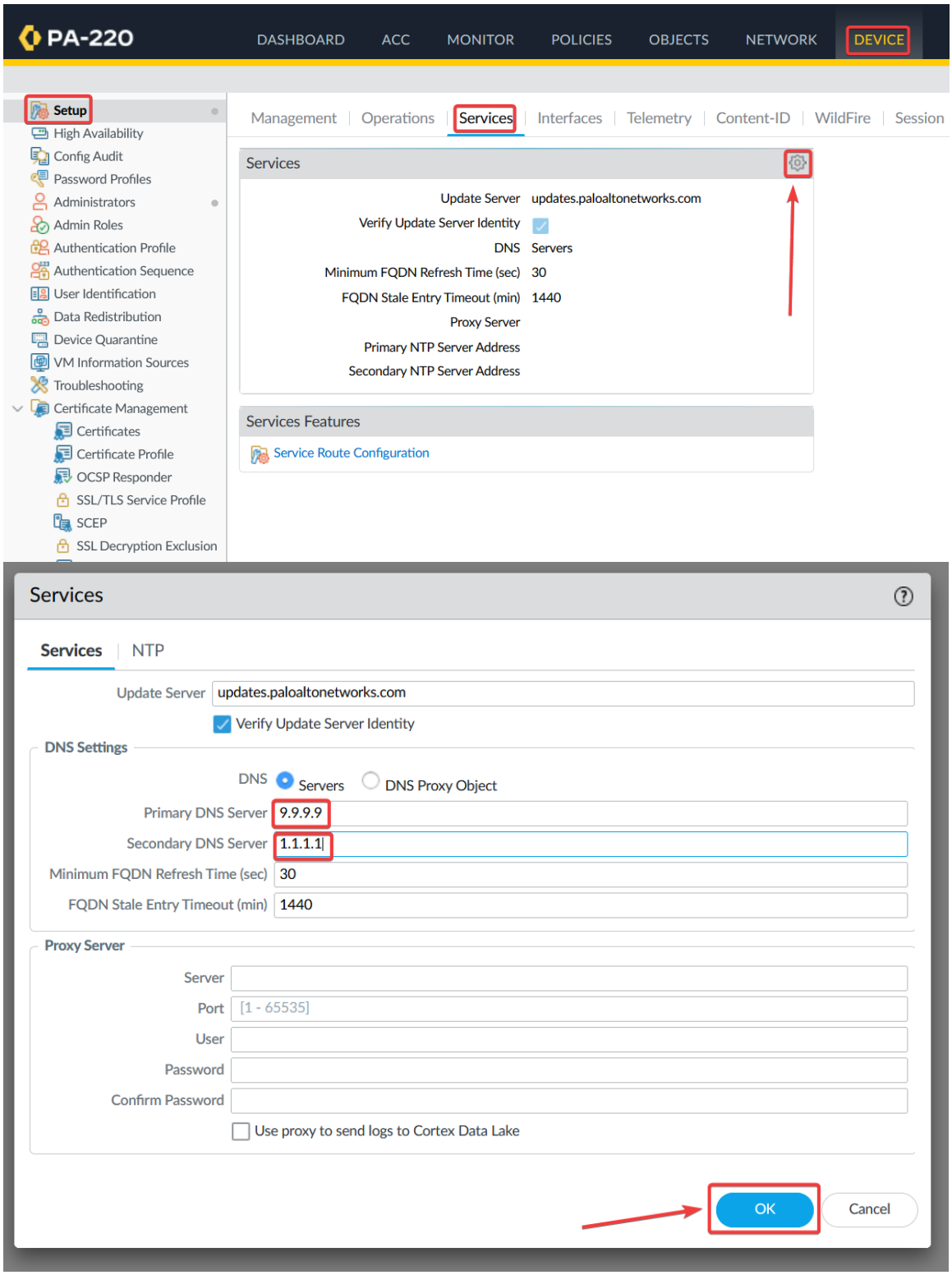
IP Address: None

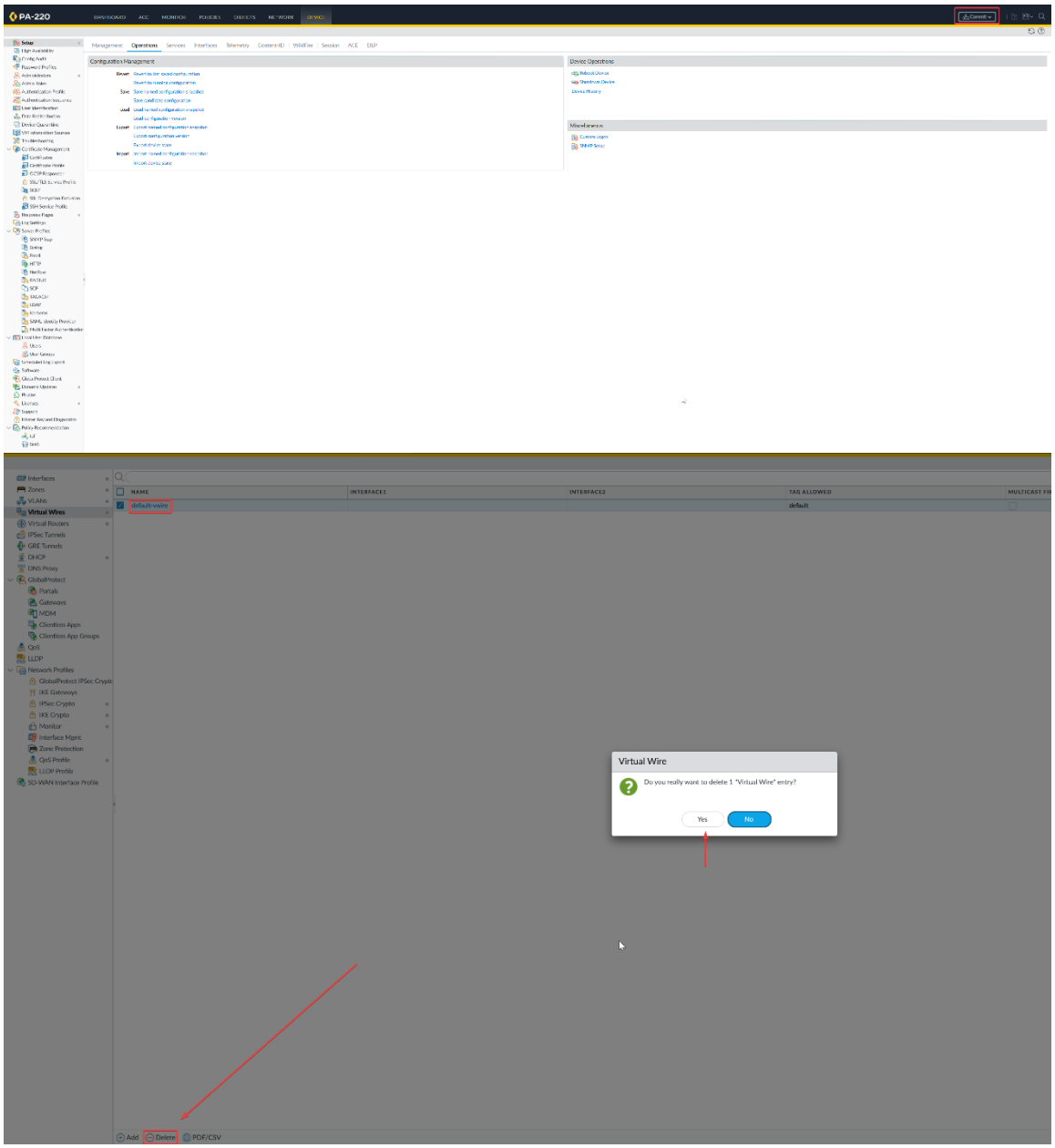
Destination Address Translation

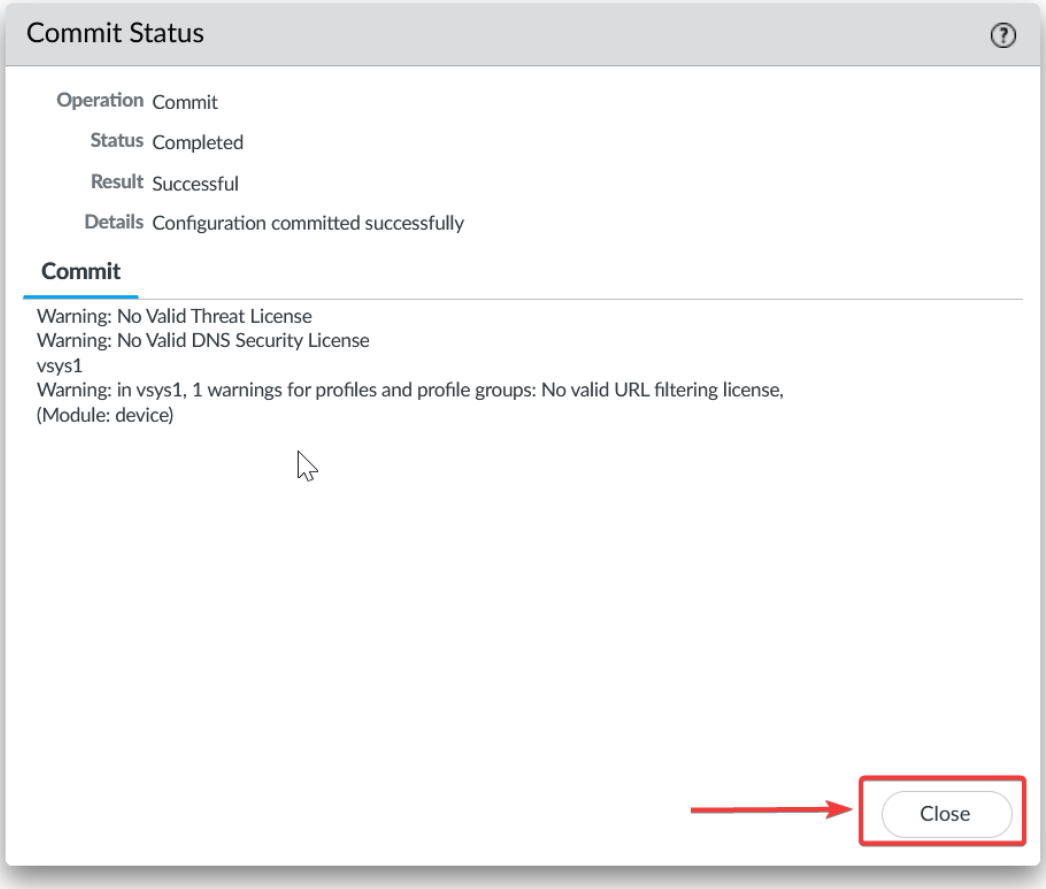
Translation Type: None

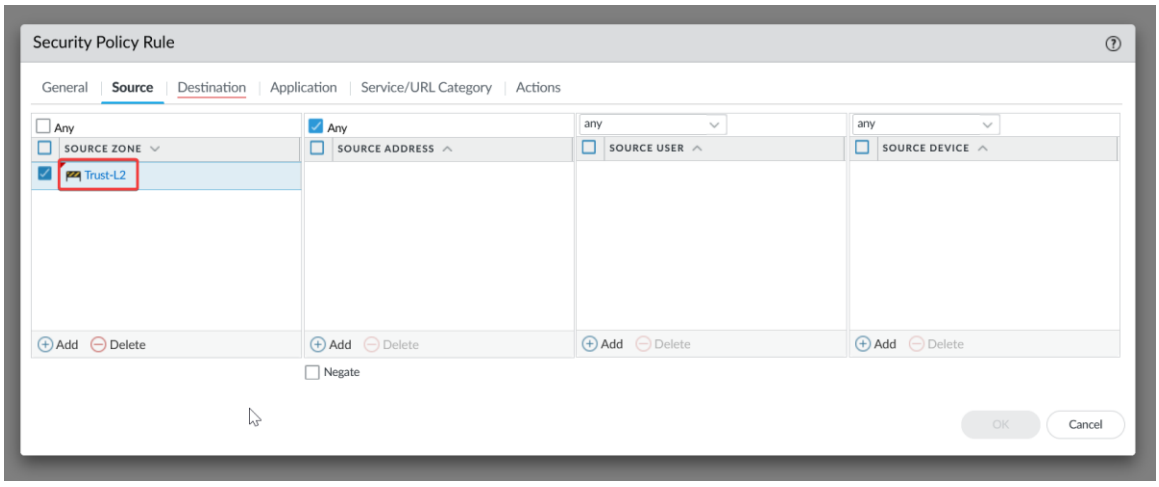
OK | Cancel

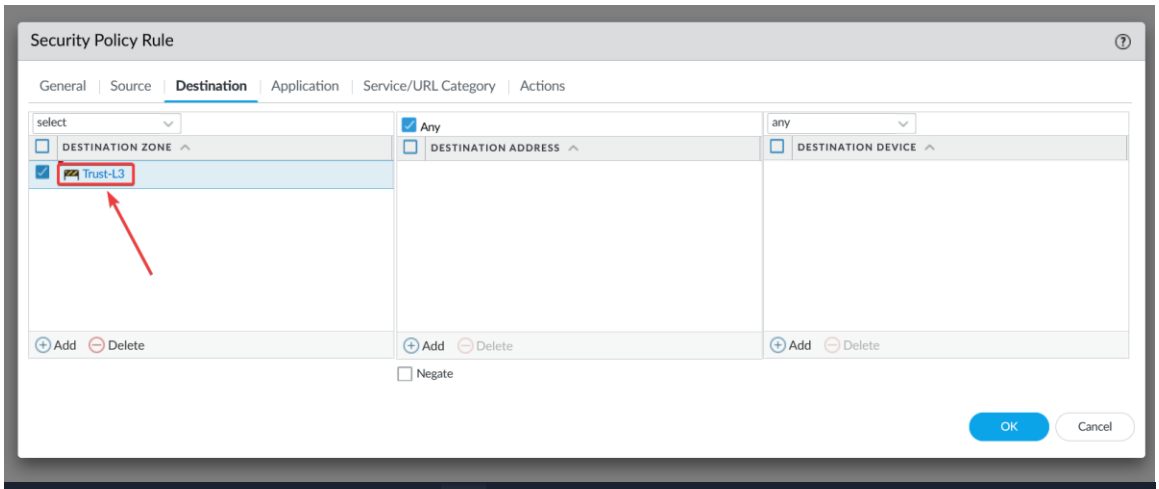

PA-220
DASHBOARD
ACC
MONITOR
POLICIES
OBJECTS
NETWORK
DEVICE
Setup
High Availability
Config Audit
Password Profiles
Administrators
Admin Roles
Authentication Profile
Authentication Sequence
User Identification
Data Redistribution
Device Quarantine
VM Information Sources
Troubleshooting
Certificate Management
Certificates
Certificate Profile
OCSP Responder
SSL/TLS Service Profile
Management
Operations
Services
Interfaces
Telemetry
Content-ID
WildFire
Sessio
INTERFACE NAME
ENABLED
Management
Management Interface Settings
IP Type
Static
DHCP Client
IP Address
192.168.1.1
Netmask
255.255.255.0
Default Gateway
192.168.1.254
IPv6 Address/Prefix Length
Default IPv6 Gateway
Speed
auto-negotiate
MTU
1500
Administrative Management Services
HTTP
HTTPS
Telnet
SSH
Network Services
HTTP OCSP
Ping
SNMP
User-ID
User-ID Syslog Listener-SSL
User-ID Syslog Listener-UDP
PERMITTED IP ADDRESSES
DESCRIPTION
Add
Delete
OK
Cancel

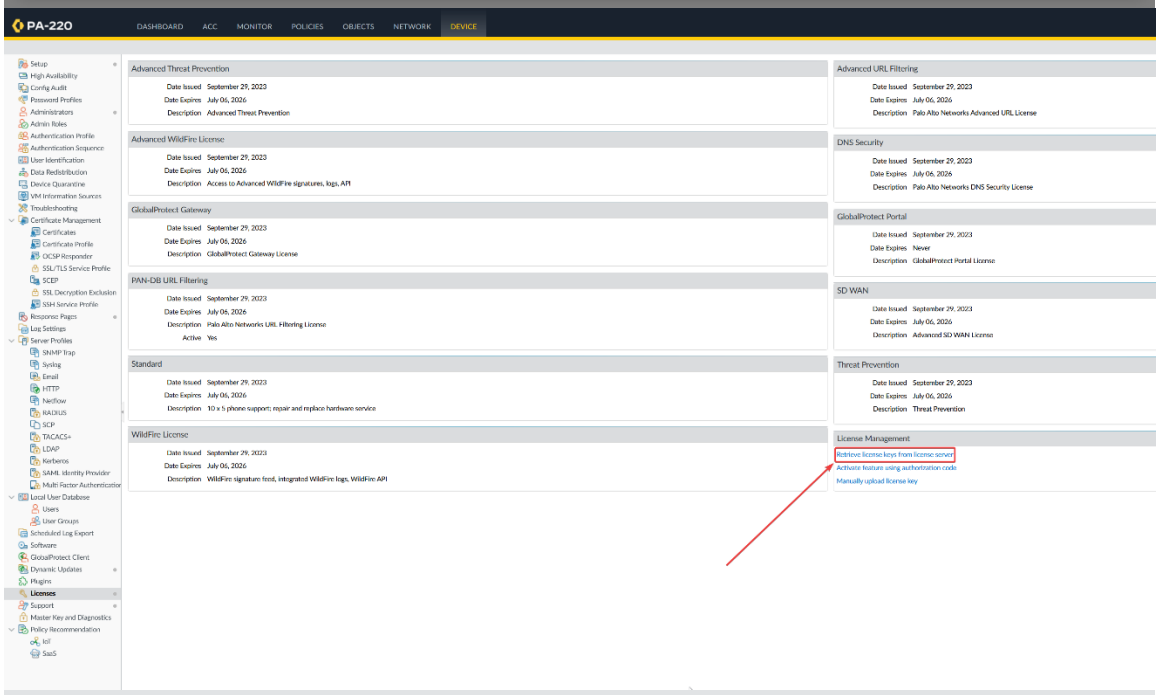

PA-220
DASHBOARD
ACC
MONITOR
POLICIES
OBJECTS
NETWORK
DEVICE
Setup
High Availability
Config Audit
Password Profiles
Administrators
Admin Roles
Authentication Profile
Authentication Sequence
User Identification
Data Redistribution
Device Quarantine
VM Information Sources
Troubleshooting
Certificate Management
Certificates
Certificate Profile
OCSP Responder
SSL/TLS Service Profile
SCEP
SSL Decryption Exclusion
Management
Operations
Services
Interfaces
Telemetry
Content-ID
WildFire
Session
Services
Update Server updates.paloaltonetworks.com
Verify Update Server Identity
DNS Servers
Minimum FQDN Refresh Time (sec) 30
FQDN Stale Entry Timeout (min) 1440
Proxy Server
Primary NTP Server Address
Secondary NTP Server Address
Services Features
Service Route Configuration
Services
Services
NTP
Update Server updates.paloaltonetworks.com
Verify Update Server Identity
DNS Settings
DNS Servers DNS Proxy Object
Primary DNS Server 9.9.9.9
Secondary DNS Server 1.1.1.1
Minimum FQDN Refresh Time (sec) 30
FQDN Stale Entry Timeout (min) 1440
Proxy Server
Server
Port [1 - 65535]
User
Password
Confirm Password
Use proxy to send logs to Cortex Data Lake
OK
Cancel

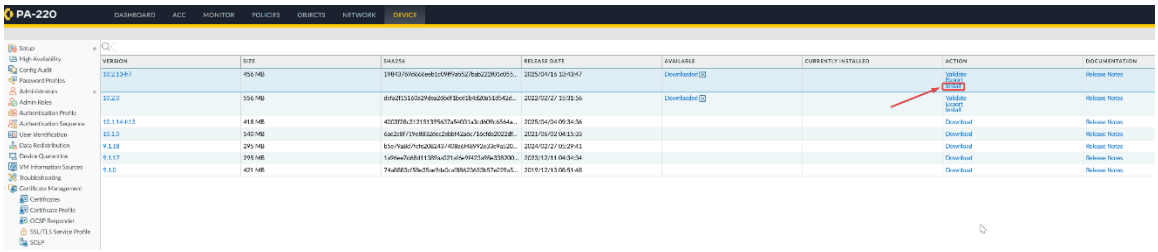

PA-220
Virtual Wires
NAME
INTERFACE1
INTERFACE2
TAG ALLOWED
default-vwire
default
Virtual Wire
Do you really want to delete 1 "Virtual Wire" entry?
Yes
No
Add
Delete
PDF/CSV


Commit Status
Operation Commit
Status Completed
Result Successful
Details Configuration committed successfully
Commit
Warning: No Valid Threat License
Warning: No Valid DNS Security License
vsys1
Warning: in vsys1, 1 warnings for profiles and profile groups: No valid URL filtering license,
(Module: device)
Close


Security Policy Rule
General
Source
Destination
Application
Service/URL Category
Actions
Any
SOURCE ZONE
Trust-L2
Any
SOURCE ADDRESS
any
SOURCE USER
any
SOURCE DEVICE
Add
Delete
Negate
OK
Cancel


Security Policy Rule
General
Source
Destination
Application
Service/URL Category
Actions
select
DESTINATION ZONE
Trust-L3
Add
Delete
Any
DESTINATION ADDRESS
Add
Delete
any
DESTINATION DEVICE
Add
Delete
Negate
OK
Cancel


PA-220
DASHBOARD
ACC
MONITOR
POLICIES
OBJECTS
NETWORK
DEVICE
Advanced Threat Prevention
Date Issued September 29, 2023
Date Expires July 06, 2026
Description Advanced Threat Prevention
Advanced WildFire License
Date Issued September 29, 2023
Date Expires July 06, 2026
Description Access to Advanced WildFire signatures, logs, API
GlobalProtect Gateway
Date Issued September 29, 2023
Date Expires July 06, 2026
Description GlobalProtect Gateway License
PAN-DB URL Filtering
Date Issued September 29, 2023
Date Expires July 06, 2026
Description Palo Alto Networks URL Filtering License
Active Yes
Standard
Date Issued September 29, 2023
Date Expires July 06, 2026
Description 10 x 5 phone support; repair and replace hardware service
WildFire License
Date Issued September 29, 2023
Date Expires July 06, 2026
Description WildFire signature feed, integrated WildFire logs, WildFire API
Advanced URL Filtering
Date Issued September 29, 2023
Date Expires July 06, 2026
Description Palo Alto Networks Advanced URL License
DNS Security
Date Issued September 29, 2023
Date Expires July 06, 2026
Description Palo Alto Networks DNS Security License
GlobalProtect Portal
Date Issued September 29, 2023
Date Expires Never
Description GlobalProtect Portal License
SD WAN
Date Issued September 29, 2023
Date Expires July 06, 2026
Description Advanced SD WAN License
Threat Prevention
Date Issued September 29, 2023
Date Expires July 06, 2026
Description Threat Prevention
License Management
Retrieve license keys from license server
Activate feature using authorization code
Manually upload license key



Configuration File

```
config {
  mgt-config {
    users {
      admin {
        phash $5$vxxcugxt$RSlQ3Mg7Q0e7hwxfuK1Hl10dDFRTrJJGxhZFuPTccSB;
        permissions {
          role-based {
            superuser yes;
          }
        }
      }
    }
    password-complexity {
      enabled yes;
      minimum-length 8;
    }
  }
  shared {
    application;
    application-group;
    service;
    service-group;
    botnet {
      configuration {
        http {
          dynamic-dns {
            enabled yes;
            threshold 5;
          }
          malware-sites {
            enabled yes;
            threshold 5;
          }
          recent-domains {
            enabled yes;
            threshold 5;
          }
          ip-domains {
            enabled yes;
            threshold 10;
          }
          executables-from-unknown-sites {
            enabled yes;
            threshold 5;
```

```
          }
        }
        other-applications {
          irc yes;
        }
        unknown-applications {
          unknown-tcp {
            destinations-per-hour 10;
            sessions-per-hour 10;
            session-length {
              maximum-bytes 100;
              minimum-bytes 50;
            }
          }
          unknown-udp {
            destinations-per-hour 10;
            sessions-per-hour 10;
            session-length {
              maximum-bytes 100;
              minimum-bytes 50;
            }
          }
        }
      }
      report {
        topn 100;
        scheduled yes;
      }
    }
  }
  devices {
    localhost.localdomain {
      network {
        interface {
          ethernet {
            ethernet1/1 {
              layer3 {
                ndp-proxy {
                  enabled no;
                }
                sdwan-link-settings {
                  upstream-nat {
                    enable no;
                    static-ip;
                  }
                  enable no;
                }
                lldp {
                  enable no;
                }
                dhcp-client;
              }
              link-state up;
            }
            ethernet1/2 {
```

```
      layer2 {
        lldp {
          enable no;
        }
      }
    }
    ethernet1/3 {
      layer2 {
        lldp {
          enable no;
        }
      }
    }
  }
  loopback {
    units;
  }
  vlan {
    units {
      vlan.10 {
        ipv6 {
          neighbor-discovery {
            router-advertisement {
              enable no;
            }
          }
        }
        ndp-proxy {
          enabled no;
        }
        adjust-tcp-mss {
          enable no;
        }
        ip {
          "IP address";
        }
      }
    }
  }
  tunnel {
    units;
  }
}
vlan {
  "Vlan Object" {
    virtual-interface {
      interface vlan.10;
    }
    interface [ ethernet1/2 ethernet1/3];
  }
}
virtual-wire;
profiles {
  monitor-profile {
    default {
```

```
        interval 3;
        threshold 5;
        action wait-recover;
      }
    }
  }
  ike {
    crypto-profiles {
      ike-crypto-profiles {
        default {
          encryption [ aes-128-cbc 3des];
          hash sha1;
          dh-group group2;
          lifetime {
            hours 8;
          }
        }
        Suite-B-GCM-128 {
          encryption aes-128-cbc;
          hash sha256;
          dh-group group19;
          lifetime {
            hours 8;
          }
        }
        Suite-B-GCM-256 {
          encryption aes-256-cbc;
          hash sha384;
          dh-group group20;
          lifetime {
            hours 8;
          }
        }
      }
      ipsec-crypto-profiles {
        default {
          esp {
            encryption [ aes-128-cbc 3des];
            authentication sha1;
          }
          dh-group group2;
          lifetime {
            hours 1;
          }
        }
        Suite-B-GCM-128 {
          esp {
            encryption aes-128-gcm;
            authentication none;
          }
          dh-group group19;
          lifetime {
            hours 1;
          }
        }
```

```
      Suite-B-GCM-256 {
        esp {
          encryption aes-256-gcm;
          authentication none;
        }
        dh-group group20;
        lifetime {
          hours 1;
        }
      }
    }
    global-protect-app-crypto-profiles {
      default {
        encryption aes-128-cbc;
        authentication sha1;
      }
    }
  }
}
qos {
  profile {
    default {
      class-bandwidth-type {
        mbps {
          class {
            class1 {
              priority real-time;
            }
            class2 {
              priority high;
            }
            class3 {
              priority high;
            }
            class4 {
              priority medium;
            }
            class5 {
              priority medium;
            }
            class6 {
              priority low;
            }
            class7 {
              priority low;
            }
            class8 {
              priority low;
            }
          }
        }
      }
    }
  }
}
```

```
virtual-router {
  default {
    protocol {
      bgp {
        enable no;
        dampening-profile {
          default {
            cutoff 1.25;
            reuse 0.5;
            max-hold-time 900;
            decay-half-life-reachable 300;
            decay-half-life-unreachable 900;
            enable yes;
          }
        }
      }
    }
    interface [ ethernet1/1 vlan.10];
  }
}
dhcp {
  interface {
    vlan.10 {
      server {
        option {
          dns {
            primary inherited;
            secondary inherited;
          }
          wins {
            primary inherited;
            secondary inherited;
          }
          nis {
            primary inherited;
            secondary inherited;
          }
          ntp {
            primary inherited;
            secondary inherited;
          }
          inheritance {
            source ethernet1/1;
          }
          lease {
            unlimited;
          }
          dns-suffix inherited;
          pop3-server inherited;
          smtp-server inherited;
          gateway 192.168.1.254;
          subnet-mask 255.255.255.0;
        }
        mode enabled;
        ip-pool 192.168.1.2-192.168.1.253;
```

```
        }
      }
    }
  }
}
deviceconfig {
  system {
    ip-address 192.168.1.1;
    netmask 255.255.255.0;
    update-server updates.paloaltonetworks.com;
    update-schedule {
      threats {
        recurring {
          weekly {
            day-of-week wednesday;
            at 01:02;
            action download-only;
          }
        }
      }
    }
    timezone US/Pacific;
    service {
      disable-telnet yes;
      disable-http yes;
    }
    hostname PA-220;
    device-telemetry {
      device-health-performance yes;
      product-usage yes;
      threat-prevention yes;
      region Americas;
    }
    type {
      static;
    }
    default-gateway 192.168.1.254;
    dns-setting {
      servers {
        primary 9.9.9.9;
        secondary 1.1.1.1;
      }
    }
  }
  setting {
    config {
      rematch yes;
    }
    management {
      hostname-type-in-syslog FQDN;
    }
  }
}
vsys {
  vsys1 {
```

```
application;
application-group;
zone {
  trust {
    network {
      virtual-wire;
    }
  }
  untrust {
    network {
      virtual-wire;
    }
  }
  Untrust-L3 {
    network {
      layer3 ethernet1/1;
    }
  }
  Trust-L3 {
    network {
      layer3 vlan.10;
    }
  }
  Trust-L2 {
    network {
      layer2 [ ethernet1/2 ethernet1/3];
    }
  }
}
service;
service-group;
schedule;
rulebase {
  security {
    rules {
      rule1 41139d30-5aa7-4680-9f01-241864e6df5e {
        from trust;
        to untrust;
        source any;
        destination any;
        service any;
        application any;
        action allow;
        log-end yes;
      }
      "Internet Outgoing" 4e6ef56b-0050-42de-951a-504e61ad9e3f {
        profile-setting {
          group Internet;
        }
        to Untrust-L3;
        from Trust-L3;
        source any;
        destination any;
        source-user any;
        category any;
```

```
                application any;
                service application-default;
                source-hip any;
                destination-hip any;
                action allow;
                description "All traffic to the internet.";
              }
            }
          }
          nat {
            rules {
              "Internet Outgoing" 56f5f543-05a6-4361-be00-14d102d5a75d {
                source-translation {
                  dynamic-ip-and-port {
                    interface-address {
                      interface ethernet1/1;
                    }
                  }
                }
                to Untrust-L3;
                from Trust-L3;
                source any;
                destination any;
                service any;
                to-interface ethernet1/1;
              }
            }
          }
        }
        import {
          network {
            interface [ ethernet1/1 ethernet1/2 ethernet1/3 vlan.10];
          }
        }
        address {
          "IP address" {
            ip-netmask 192.168.1.254/24;
          }
        }
        profile-group {
          Internet {
            virus default;
            spyware strict;
            vulnerability strict;
            url-filtering default;
          }
        }
      }
    }
  }
}

admin@PA-220> show config running
+ xpath    xpath of the node to retrieve
```

```
  |        Pipe through a command
  <Enter>  Finish input

admin@PA-220> set cli config-output-format set
admin@PA-220> set cli config-output-format set
  <Enter>  Finish input

admin@PA-220> show config running

config {
  mgt-config {
    users {
      admin {
        phash $5$vxxcugxt$RSlQ3Mg7Q0e7hwxfuK1Hl10dDFRTrJJGxhZFuPTccSB;
        permissions {
          role-based {
            superuser yes;
          }
        }
      }
    }
    password-complexity {
      enabled yes;
      minimum-length 8;
    }
  }
  shared {
    application;
    application-group;
    service;
    service-group;
    botnet {
      configuration {
        http {
          dynamic-dns {
            enabled yes;
            threshold 5;
          }
          malware-sites {
            enabled yes;
            threshold 5;
          }
          recent-domains {
            enabled yes;
            threshold 5;
          }
          ip-domains {
            enabled yes;
            threshold 10;
          }
          executables-from-unknown-sites {
            enabled yes;
            threshold 5;
          }
        }
```

```
      other-applications {
        irc yes;
      }
      unknown-applications {
        unknown-tcp {
          destinations-per-hour 10;
          sessions-per-hour 10;
          session-length {
            maximum-bytes 100;
            minimum-bytes 50;
          }
        }
        unknown-udp {
          destinations-per-hour 10;
          sessions-per-hour 10;
          session-length {
            maximum-bytes 100;
            minimum-bytes 50;
          }
        }
      }
    }
    report {
      topn 100;
      scheduled yes;
    }
  }
}
devices {
  localhost.localdomain {
    network {
      interface {
        ethernet {
          ethernet1/1 {
            layer3 {
              ndp-proxy {
                enabled no;
              }
              sdwan-link-settings {
                upstream-nat {
                  enable no;
                  static-ip;
                }
                enable no;
              }
              lldp {
                enable no;
              }
              dhcp-client;
            }
            link-state up;
          }
          ethernet1/2 {
            layer2 {
              lldp {
```

```
            enable no;
          }
        }
      }
      ethernet1/3 {
        layer2 {
          lldp {
            enable no;
          }
        }
      }
    }
    loopback {
      units;
    }
    vlan {
      units {
        vlan.10 {
          ipv6 {
            neighbor-discovery {
              router-advertisement {
                enable no;
              }
            }
          }
          ndp-proxy {
            enabled no;
          }
          adjust-tcp-mss {
            enable no;
          }
          ip {
            "IP address";
          }
        }
      }
    }
    tunnel {
      units;
    }
  }
  vlan {
    "Vlan Object" {
      virtual-interface {
        interface vlan.10;
      }
      interface [ ethernet1/2 ethernet1/3];
    }
  }
  virtual-wire;
  profiles {
    monitor-profile {
      default {
        interval 3;
        threshold 5;
```

```
      action wait-recover;
    }
  }
}
ike {
  crypto-profiles {
    ike-crypto-profiles {
      default {
        encryption [ aes-128-cbc 3des];
        hash sha1;
        dh-group group2;
        lifetime {
          hours 8;
        }
      }
      Suite-B-GCM-128 {
        encryption aes-128-cbc;
        hash sha256;
        dh-group group19;
        lifetime {
          hours 8;
        }
      }
      Suite-B-GCM-256 {
        encryption aes-256-cbc;
        hash sha384;
        dh-group group20;
        lifetime {
          hours 8;
        }
      }
    }
    ipsec-crypto-profiles {
      default {
        esp {
          encryption [ aes-128-cbc 3des];
          authentication sha1;
        }
        dh-group group2;
        lifetime {
          hours 1;
        }
      }
      Suite-B-GCM-128 {
        esp {
          encryption aes-128-gcm;
          authentication none;
        }
        dh-group group19;
        lifetime {
          hours 1;
        }
      }
      Suite-B-GCM-256 {
        esp {
```

```
            encryption aes-256-gcm;
            authentication none;
          }
          dh-group group20;
          lifetime {
            hours 1;
          }
        }
      }
      global-protect-app-crypto-profiles {
        default {
          encryption aes-128-cbc;
          authentication sha1;
        }
      }
    }
  }
  qos {
    profile {
      default {
        class-bandwidth-type {
          mbps {
            class {
              class1 {
                priority real-time;
              }
              class2 {
                priority high;
              }
              class3 {
                priority high;
              }
              class4 {
                priority medium;
              }
              class5 {
                priority medium;
              }
              class6 {
                priority low;
              }
              class7 {
                priority low;
              }
              class8 {
                priority low;
              }
            }
          }
        }
      }
    }
  }
  virtual-router {
    default {
```

```
    protocol {
      bgp {
        enable no;
        dampening-profile {
          default {
            cutoff 1.25;
            reuse 0.5;
            max-hold-time 900;
            decay-half-life-reachable 300;
            decay-half-life-unreachable 900;
            enable yes;
          }
        }
      }
    }
    interface [ ethernet1/1 vlan.10];
  }
}
dhcp {
  interface {
    vlan.10 {
      server {
        option {
          dns {
            primary inherited;
            secondary inherited;
          }
          wins {
            primary inherited;
            secondary inherited;
          }
          nis {
            primary inherited;
            secondary inherited;
          }
          ntp {
            primary inherited;
            secondary inherited;
          }
          inheritance {
            source ethernet1/1;
          }
          lease {
            unlimited;
          }
          dns-suffix inherited;
          pop3-server inherited;
          smtp-server inherited;
          gateway 192.168.1.254;
          subnet-mask 255.255.255.0;
        }
        mode enabled;
        ip-pool 192.168.1.2-192.168.1.253;
      }
    }
```

```
    }
  }
}
deviceconfig {
  system {
    ip-address 192.168.1.1;
    netmask 255.255.255.0;
    update-server updates.paloaltonetworks.com;
    update-schedule {
      threats {
        recurring {
          weekly {
            day-of-week wednesday;
            at 01:02;
            action download-only;
          }
        }
      }
    }
    timezone US/Pacific;
    service {
      disable-telnet yes;
      disable-http yes;
    }
    hostname PA-220;
    device-telemetry {
      device-health-performance yes;
      product-usage yes;
      threat-prevention yes;
      region Americas;
    }
    type {
      static;
    }
    default-gateway 192.168.1.254;
    dns-setting {
      servers {
        primary 9.9.9.9;
        secondary 1.1.1.1;
      }
    }
  }
  setting {
    config {
      rematch yes;
    }
    management {
      hostname-type-in-syslog FQDN;
    }
  }
}
vsys {
  vsys1 {
    application;
    application-group;
```

```
zone {
  trust {
    network {
      virtual-wire;
    }
  }
  untrust {
    network {
      virtual-wire;
    }
  }
  Untrust-L3 {
    network {
      layer3 ethernet1/1;
    }
  }
  Trust-L3 {
    network {
      layer3 vlan.10;
    }
  }
  Trust-L2 {
    network {
      layer2 [ ethernet1/2 ethernet1/3];
    }
  }
}
service;
service-group;
schedule;
rulebase {
  security {
    rules {
      rule1 41139d30-5aa7-4680-9f01-241864e6df5e {
        from trust;
        to untrust;
        source any;
        destination any;
        service any;
        application any;
        action allow;
        log-end yes;
      }
      "Internet Outgoing" 4e6ef56b-0050-42de-951a-504e61ad9e3f {
        profile-setting {
          group Internet;
        }
        to Untrust-L3;
        from Trust-L3;
        source any;
        destination any;
        source-user any;
        category any;
        application any;
        service application-default;
```

```
                source-hip any;
                destination-hip any;
                action allow;
                description "All traffic to the internet.";
              }
            }
          }
          nat {
            rules {
              "Internet Outgoing" 56f5f543-05a6-4361-be00-14d102d5a75d {
                source-translation {
                  dynamic-ip-and-port {
                    interface-address {
                      interface ethernet1/1;
                    }
                  }
                }
                to Untrust-L3;
                from Trust-L3;
                source any;
                destination any;
                service any;
                to-interface ethernet1/1;
              }
            }
          }
        }
        import {
          network {
            interface [ ethernet1/1 ethernet1/2 ethernet1/3 vlan.10];
          }
        }
        address {
          "IP address" {
            ip-netmask 192.168.1.254/24;
          }
        }
        profile-group {
          Internet {
            virus default;
            spyware strict;
            vulnerability strict;
            url-filtering default;
          }
        }
      }
    }
  }
}
```